

Acute Ischemic Stroke Treatments - Medical Clinical Policy Bulletins

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses selected treatments for acute ischemic stroke.

1. Medical Necessity

Aetna considers the following medically necessary:

1. Endovascular therapy with a retrievable stent (e.g., Solitaire FR (Flow Restoration stent retriever, Trevo retriever) for persons with acute ischemic stroke who have occlusion of the anterior circulation (i.e., middle cerebral artery trunk or its branches or internal carotid artery) and evidence of salvageable tissue on imaging, where retrieval is performed within 12 hours after the onset of stroke;
2. Intra-arterial infusion of spasmolytics (e.g., papaverine) or calcium-channel blockers (e.g., nicardapine) into the intracranial arteries for the treatment of medically refractory symptomatic delayed cerebral ischemia (cerebral vasospasm) after aneurysmal subarachnoid hemorrhage.
Infusion of spasmolytics or calcium channel blockers into the intracranial arteries is considered experimental, investigational, or unproven for all other indications.

2. Experimental, Investigational, or Unproven

Aetna considers the following experimental, investigational, or unproven (not an all-inclusive list):

- Acute stenting and concomitant tirofiban administration for the treatment of acute ischemic stroke (AIS)
- Cerebrolysin for the treatment of AIS
- Defibrinogen therapy (ancrod, batroxobin)
- Erythropoietin for the treatment of AIS
- Exosome-based therapies for the treatment of AIS
- Glycoprotein IIb/IIIa antagonists (e.g., tirofiban) for the treatment of AIS

- Governor vessel acupuncture for the treatment of AIS
- Hypothermia for the treatment of AIS
- Hypoxia-induced neuroprotection (intermittent hypoxic exposures or intermittent hypoxic-hyperoxic exposures)
- Mesenchymal stem cells for enhancement of ischemic stroke recovery
- Microbubbles combined with ultrasound sonothrombolysis
- Minocycline
- Nanoparticle-based therapies
- Normo-baric oxygen therapy
- Prophylactic infusion of spasmolytics or calcium-channel blockers into the intracranial arteries after aneurysmal subarachnoid hemorrhage. Infusion of spasmolytics or calcium channel blockers into the intracranial arteries for all other indications
- Sphenopalatine ganglion stimulation for the treatment of AIS
- Statins (e.g., simvastatin)
- Tirilazad for the treatment of subarachnoid hemorrhage
- Transcranial ultrasound (e.g., the CLOTBUST-HF device), alone or in combination with tissue plasminogen activator, for the treatment of AIS
- Transdermal glyceryl trinitrate for the treatment of AIS
- Ultrasound-enhanced thrombolysis for the treatment of AIS
- Vagus nerve stimulation for improvement of myocardial atrophy after AIS.

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
<i>Intra-arterial infusion of spasmolytics or calcium antagonists hypothermia or transcranial ultrasound - no specific code:</i>	
37184	Primary percutaneous transluminal mechanical thrombectomy, noncoronary, arterial or arterial bypass graft, including fluoroscopic guidance and intraprocedural pharmacological thrombolytic injection(s); initial vessel
37185	second and all subsequent vessel(s) within the same vascular family (List separately in addition to code for primary mechanical thrombectomy procedure)
61645	Percutaneous arterial transluminal mechanical thrombectomy and/or infusion for thrombolysis, intracranial, any method, including diagnostic angiography, fluoroscopic guidance, catheter placement, and intraprocedural pharmacological thrombolytic injection(s)
CPT codes not covered for indications listed in the CPB:	

Code	Code Description
<i>Acute stenting, hypothermia or transcranial ultrasound device (e.g., the CLOTBUST-HF device), defibrinogen therapy, microbubbles combined with ultrasound sonothrombolysis, sphenopalatine ganglion stimulation, normo-baric oxygen therapy, exosome-based therapies for the treatment of acute ischemic stroke, Hypoxia-induced neuroprotection (intermittent hypoxic exposure) or intermittent hypoxic-hyperoxic exposures, nanoparticle-based therapies - no specific code</i>	
38206	Blood-derived hematopoietic progenitor cell harvesting for transplantation, per collection; autologous
38232	Bone marrow harvesting for transplantation; autologous
38241	Hematopoietic progenitor cell (HPC); autologous transplantation
61885	Insertion or replacement of a cranial neurostimulator pulse generator or receiver
61886	Insertion or replacement of a cranial neurostimulator pulse generator or receiver with connection to two or more electrode arrays
61888	Revision or removal of cranial neurostimulator pulse generator or receiver
64568	Open implantation of cranial nerve (eg, vagus nerve) neurostimulator electrode array and pulse generator
64569	Revision or replacement of cranial nerve (eg, vagus nerve) neurostimulator electrode array, including connection to existing pulse generator
97810 – 97814	Acupuncture [Governor vessel]
HCPCS codes not covered for indications listed in the CPB:	
<i>Cerebrolysin, Transdermal glyceryl trinitrate, Statins (e.g. simvastatin) - no specific code:</i>	
C1767	Generator, neurostimulator (implantable), nonrechargeable
C1778	Lead, neurostimulator (implantable)
C1816	Receiver and/or transmitter, neurostimulator (implantable)
C1827	Generator, neurostimulator (implantable), non-rechargeable, with implantable stimulation lead and external paired stimulation controller
C1883	Adaptor/ extension, pacing lead or neurostimulator lead (implantable)
E0735	Non-invasive vagus nerve stimulator
E0770	Functional electrical stimulator, transcutaneous stimulation of nerve and/or muscle groups, any type, complete system, not otherwise specified
J0885	Injection, epoetin alfa, (for non-esrd use), 1000 units
J0887	Injection, epoetin beta, 1 microgram, (for esrd on dialysis)
J0888	Injection, epoetin beta, 1 microgram, (for non esrd use)
J2265	Injection, minocycline HCl, 1 mg
J3246	Injection, tirofiban HCl, 0.25 mg

Code	Code Description
L8680	Implantable neurostimulator electrode, each
L8681	Patient programmer (external) for use with implantable programmable neurostimulator pulse generator, replacement only
L8682	Implantable neurostimulator radiofrequency receiver
L8683	Radiofrequency transmitter (external) for use with implantable neurostimulator radiofrequency receiver
L8685	Implantable neurostimulator pulse generator, single array, rechargeable, includes extension
L8686	Implantable neurostimulator pulse generator, single array, non-rechargeable, includes extension
L8687	Implantable neurostimulator pulse generator, dual array, rechargeable, includes extension
L8688	Implantable neurostimulator pulse generator, dual array, non-rechargeable, includes extension
L8689	External recharging system for battery (internal) for use with implanted neurostimulator, replacement only
L8695	External recharging system for battery (external) for use with implantable neurostimulator, replacement only
Q4081	Injection, epoetin alfa, 100 units (for esrd on dialysis) epoetin alfa
Other HCPCS codes related to the CPB:	
C1757	Catheter, thrombectomy/embolectomy
C1876	Stent, non-coated/non-covered, with delivery system
C1884	Embolization protective system
C1887	Catheter, guiding (may include infusion/perfusion capability)
ICD-10 codes covered if selection criteria are met:	
I63.00 - I63.9	Cerebral infarction [acute ischemic stroke]
I67.82	Cerebral ischemia [medically refractory symptomatic delayed cerebral ischemia]
I67.841 - I67.848	Cerebral vasospasm and vasoconstriction [medically refractory symptomatic delayed cerebral ischemia]
ICD-10 codes not covered for indications listed in the CPB:	
I51.5	Myocardial degeneration

Background

Acute ischemic stroke (AIS) is among the leading causes of death and disability in developed countries. Traditional treatment entails the use of anti-coagulants and/or

aspirin. Within the appropriate time-window, various endovascular approaches have been employed to manage patients with AIS. Endovascular therapy comprises a number of pharmacological and mechanical procedures. Intravenous (IV) thrombolysis including the use of tissue plasminogen activator (tPA) is an accepted treatment for AIS administered within 3 hours of onset. Mechanical procedures including the use of various micro-guidewires, micro-snare, and retrievers (e.g., the mechanical embolus removal in cerebral ischemia [MERC] device, the L5 Retriever, and the Penumbra System) offer the promise of effective treatment for patients in whom pharmacological thrombolysis is contraindicated or might be ineffective. Although earlier devices had shown mixed results, better results have been obtained with modern stent retrievers combined with advanced imaging to select patients with viable tissue for rescue.

In a prospective, observational, cohort study, González et al (2007) evaluated the safety and effectiveness of thrombus extraction using a micro-snare in patients with AIS. Consecutive patients with AIS (less than 6 hours of ischemia for anterior circulation and less than 24 hours for posterior circulation) who had been previously excluded from IV tPA thrombolysis were included and followed-up for 3 months. Mechanical embolectomy with a micro-snare of 2 to 4 mm was undertaken as the first treatment. Low-dose intra-arterial (IA) thrombolysis or angioplasty was used if needed. Thrombolysis in Myocardial Infarction (TIMI) grade and modified Rankin Scale (mRS) score were used to evaluate vessel re-canalization and clinical effectiveness, respectively. A total of 9 patients (mean age of 55 years, range of 17 to 69 years) were included. Their basal mean National Institutes of Health Stroke Scale (NIHSS) score was 16 (range of 12 to 24). In 7 out of the 9 patients (77.8 %) the clot was removed, giving a TIMI grade of 3 (n = 4) and TIMI grade 2 (n = 3). Occlusion sites were: middle cerebral artery (MCA, n = 4), basilar artery (n = 2) and anterior cerebral artery plus MCA (n = 1). The mean time for re-canalization from the start of the procedure was 50 mins (range of 50 to 75 mins). At 3 months, the mRS score was 0 (n = 2) and 3 to 4 (n = 3; 2 patients died). The authors concluded that the micro-snare is a safe procedure for mechanical thrombectomy with a good re-canalization rate. Moreover, they stated that further studies are needed to determine the role of the micro-snare in the treatment of AIS.

Bose and associates (2008) noted that recent findings suggested that mechanical thrombectomy may have potential as a treatment for AIS. In a prospective, single-arm, trial, these investigators evaluated the safety and performance of the Penumbra System (PS), a novel mechanical device designed to reduce clot burden in acute stroke due to large-vessel occlusive disease. Patients with an acute neurological deficit consistent with acute stroke, presenting within 8 hours of symptom onset and an angiographically verified occlusion (TIMI grade 0 or 1) of a treatable intra-cranial vessel were enrolled in this study. The primary end point was re-vascularization of the target vessel to TIMI grade 2 or 3. Secondary end points were the proportion of subjects who achieved a mRS score of 2 or less, or a 4-point improvement on the NIHSS score at 30-day follow-up, as well as all-cause mortality. A total of 23 subjects were enrolled, and 21 target vessels were treated in 20 subjects by the PS. At baseline, mean age was 60 years, mean mRS score was 4.6, and mean NIHSS score was 21. Post-procedure, all 21 of the treated vessels (100 %) were successfully re-vascularized by the PS to TIMI 2 or 3. At 30-day follow-up, 9 subjects (45 %) had a 4-point or more NIHSS improvement or an mRS of 2 or less. The all-cause mortality rate was 45 % (9 of 20), which is lower than expected in this severe stroke cohort, where 70 % of the subjects at baseline had either an NIHSS score of more than 20 or a basilar occlusion. The authors concluded that early clinical experience suggested that the Penumbra System allows re-vascularization in certain subjects experiencing AIS.

Smith (2007) noted that 2 major randomized trials of IV thrombolytic therapy have established clear clinical benefit, especially for strokes caused by small-vessel occlusion. Ischemic stroke caused by large-vessel intra-cranial occlusion carries higher morbidity, however, and IV thrombolytics are less capable of opening these large vessels. This observation makes a case for delivering thrombolytics directly into the clot, or simply removing the clot mechanically. Intra-arterial thrombolytic drugs have been shown to be effective for treating MCA occlusions in a major randomized trial. In the past two years, a family of mechanical thrombectomy catheters designed to remove rather than dissolve the offending clot has received Food and Drug Administration (FDA) clearance (e.g., the MERC] device). Such devices offer alternative therapy to patients who can not receive thrombolytics, and can also be used in combination with thrombolytics to safely restore cerebral perfusion. Thomassen and Bakke (2007) stated that mechanical embolectomy works well on large-volume proximal occlusions for which there was previously no effective treatment. Early safety trials are promising, effectiveness in terms of re-

canalization is substantial, and both safety and efficacy is expected to improve with further advances in technology. The authors concluded that IV thrombolysis with tPA revolutionized acute stroke treatment a decade ago; endovascular therapy offers the promise of a second revolution, expanding the number of patients eligible and the time-window open for specific stroke treatment.

In a prospective, non-randomized, multi-center trial, Smith et al (2005) examined the safety and effectiveness of the Merci retriever to open occluded intra-cranial large vessels within 8 hours of the onset of stroke symptoms. All patients were ineligible for IV tPA. Primary outcomes were re-canalization and safety; secondary outcomes were neurological outcome at 90 days in re-canalized versus non re-canalized patients. Re-canalization was achieved in 46 % (69/151) of patients on intention-to-treat analysis, and in 48 % (68/141) of patients in whom the device was deployed. This rate is significantly higher than that expected using an historical control (Prolyse in Acute Cerebral Thromboembolism II [PROACT II] trial) of 18 % ($p < 0.0001$). Clinically significant procedural complications occurred in 10 of 141 (7.1 %) patients. Symptomatic intra-cranial hemorrhages was observed in 11 of 141 (7.8 %) patients. Good neurological outcomes (mRS score less than or equal to 2) were more frequent at 90 days in patients with successful re-canalization compared with patients with unsuccessful re-canalization (46 % versus 10 %; relative risk [RR], 4.4; 95 % confidence interval [CI]: 2.1 to 9.3; $p < 0.0001$), and mortality was less (32 % versus 54 %; RR, 0.59; 95 % CI: 0.39 to 0.89; $p = 0.01$). The authors concluded that the MERCI retriever can significantly restore vascular patency during AIS within 8 hours of stroke symptom onset and provides an alternative intervention for patients who are otherwise ineligible for thrombolytics. However, there are various drawbacks with the findings of this study (Oliveira-Filho et al, 2009):

- Neurological outcome was a secondary end point in the MERCI trial, and there was no evidence that treated patients had improved outcome at 90 days compared with PROACT II historical controls.
- On an intention-to-treat basis, patients in the MERCI trial had a higher re-canalization rate (TIMI 2 or 3 flow as reported by the local operator) than the spontaneous re-canalization rate for historical controls (46 % versus 18 %). However, while the device is intended to restore blood flow by removing thrombus, MERCI has been criticized because there was no confirmation that blood flow was actually restored by embolectomy as opposed to clot disruption and ensuing distal embolization. In addition, the study lacked a specific definition of TIMI re-canalization.
- Overall mortality by intention-to-treat analysis was higher in MERCI than in PROACT II historical controls (44 % versus 27 %), which may be due, at least in part, to older patients with slightly more severe strokes at baseline in MERCI compared with PROACT II. Although MERCI patients had a higher rate of symptomatic intra-cranial hemorrhage than the historical controls (8 % versus 2 %, respectively), the 8 % rate in MERCI is similar to the 6.4 % rate seen with IV tPA treatment in the National Institute of Neurological Disorders and Stroke trial.
- Procedural complications in MERCI, including embolization, dissection, subarachnoid hemorrhage (SAH), vessel perforation, and groin hemorrhages, occurred in 13 % and were considered clinically significant in 7 %.

The Canadian Coordinating Office for Health Technology Assessment (2005) stated that "[t]he Merci Retrieval System is another step in the development of new approaches for treating ischemic stroke. It provides an addition option when thrombolysis has failed or is inappropriate. Use of the device requires a high skill level. Data on the effectiveness of the device are limited; it is expected that the results of ongoing trials will provide further information".

An assessment prepared for the Agency for Healthcare Research and Quality concluded that future well-designed studies investigating mechanical thrombus disruption to reduce stroke-related mortality and disability are needed (Sharma et al, 2005). In addition, the California Technology Assessment Forum (2007) concluded that the use of the Merci retriever for the emergent treatment of AIS does not meet CTAF criteria.

Smith and co-workers (2008) stated that first-generation MERCI devices achieved re-canalization rates of 48 % and, when coupled with IA thrombolytic drugs, re-canalization rates of 60 % have been reported; and enhancements in embolectomy device design may improve re-canalization rates. The Multi-MERCI was an international, multi-center, prospective, single-arm trial of thrombectomy in patients with large-vessel stroke treated within 8 hours of symptom onset. Patients with persistent large-vessel occlusion

after IV tPA treatment were included. Once the newer generation (L5 Retriever) device became available, investigators were instructed to use the L5 Retriever to open vessels and could subsequently use older generation devices and/or IA tPA. Primary outcome was re-canalization of the target vessel. A total of 164 patients received thrombectomy and 131 were initially treated with the L5 Retriever. Mean age $\pm$ SD was 68 $\pm$ 16 years, and baseline median (inter-quartile range) NIHSS score was 19 (15 to 23). Treatment with the L5 Retriever resulted in successful re-canalization in 75 of 131 (57.3 %) treatable vessels and in 91 of 131 (69.5 %) after adjunctive therapy (IA tPA, mechanical). Overall, favorable clinical outcomes (mRS 0 to 2) occurred in 36 % and mortality was 34 %; both outcomes were significantly related to vascular re-canalization. Symptomatic intra-cerebral hemorrhage occurred in 16 patients (9.8 %); 4 (2.4 %) of these were parenchymal hematoma type II. Clinically significant procedural complications occurred in 9 (5.5 %) patients. The authors concluded that higher rates of re-canalization were associated with a newer generation thrombectomy device compared with first-generation devices, but these differences did not achieve statistical significance. Mortality trended lower and the proportion of good clinical outcomes trended higher, consistent with better re-canalization.

Kobayashi et al (2008) reported of the first 2 cases of AIS treated with the MERCI device at the authors' department. One did not meet the inclusion criteria for systemic thrombolysis, and the second did not improve despite r-tPA treatment. In both cases, improvement of flow in the MCA was achieved and moderate neurological improvement was observed at 3-month follow-up. The authors stated that more controlled trials are needed to establish the utility of mechanical embolectomy in the treatment of stroke.

Josephson et al (2009) noted that IA thrombolysis and mechanical embolectomy have been studied for endovascular treatment of stroke. The MERCI and Multi-MERCI trials of mechanical embolectomy with or without adjuvant IA thrombolysis demonstrated effective re-canalization, but with a higher mortality compared with control patients in the PROACT II trial of IA thrombolysis. Differences in trial design may account for this mortality difference. These investigators identified patients in the MERCI and Multi-MERCI trials who would have been eligible for PROACT II. Rates of good outcome (mRS less than or equal to 2) and mortality at 90 days were compared, adjusting for differences in baseline NIHSS score and age. A total of 68 patients enrolled in MERCI and 81 enrolled in Multi-MERCI were eligible for PROACT II. In both unadjusted and adjusted analyses, PROACT II-eligible embolectomy patients showed a trend toward better clinical outcomes compared to the PROACT II control-arm (adjusted, MERCI 35.4 % [p = ns], Multi-MERCI 42.8 % [p = 0.048], PROACT II control, 25.4 %). In both unadjusted and adjusted analyses, mortality rates did not significantly differ between embolectomy patients and PROACT II control patients (adjusted analysis, MERCI 29.1 %, Multi-MERCI 18.0 %, PROACT II control, 27.1 %). Compared with the PROACT II treatment group, embolectomy groups showed similar rates of good outcome and mortality. The authors concluded that differences in mortality and proportion of good outcome between the MERCI/Multi-MERCI trials and the PROACT II trial are explained by differences in study design and baseline characteristics of patients. Mechanical embolectomy and IA thrombolysis may each be reasonable strategies for AIS; a randomized trial is necessary to confirm these results.

Fields et al (2010) stated that AIS is the leading cause of severe disability and the third leading cause of death in the United States. Intravenous tissue plasminogen activator (IV tPA) remains the most widely advocated treatment, but this therapy is limited by a narrow time window (less than 4.5 hours after stroke onset), exclusion of patients with coagulopathy and re-canalization rates of less than 50 %. As a result, only 5 % of acute stroke patients are treated with IV tPA. Endovascular mechanical thrombectomy may be employed, either as a stand-alone therapy or as an adjunct to IV tPA, and has several potential advantages, including a wider time window (up to 8 hours), the capacity for use in coagulopathic patients and higher re-canalization rates (up to 82 %). Nonetheless, mechanical thrombectomy has engendered controversy because no randomized trials have yet been performed to support its use. The authors concluded that the results of ongoing trials are needed to ascertain the patient populations most likely to benefit from this therapy.

Sugiura and colleagues (2008) examined the safety and effectiveness of combined IV recombinant-tPA (r-tPA) and simultaneous endovascular therapy (ET) as primary rather than rescue therapy for hyper-acute MCA occlusion. A total of 29 patients eligible for IV r-tPA, who were diagnosed as having MCA (M1 or M2) occlusion within 3 hours of onset, underwent thrombolysis. In the combined group, patients were treated by IV r-tPA (0.6 mg/kg for 60 mins) and simultaneous ET (intra-arterial r-tPA, mechanical thrombus

disruption with micro-guidewire, and balloon angioplasty) initiated as soon as possible. In the IV group, patients were treated by IV r-tPA only. The improvement of the NIHSS score at 24 hours was 11.0 +/- 4.8 in the combined group versus 5.0 +/- 4.3 in the IV group ($p < 0.001$). In the combined group, successful re-canalization was observed in 14 (88 %) of 16 patients with no symptomatic intra-cranial hemorrhage, and 10 (63 %) of 16 patients had favorable outcomes (mRS 0, 1) at 3 months. The authors concluded that aggressive combined therapy with IV r-tPA and simultaneous ET markedly improved the clinical outcome of hyper-acute MCA occlusion without significant adverse effect. Moreover, they stated that additional randomized study is needed to confirm these findings.

Stead et al (2008) performed a systematic review and meta-analysis of mechanical thrombectomy in the treatment of ischemic stroke and assessed factors for technical and clinical success and survival. These researchers searched the literature using Medline and Embase for January 1, 2000 through March 1, 2006. Studies were limited to those in human beings; there were no language or study design restrictions. Validity assessment was performed using the Newcastle-Ottawa Scale. The pooled cohort was compared with a historical cohort matched for sex, age, and NIHSS. The search yielded 114 publications. Two authors determined inclusibility (inter-rater agreement, kappa = 0.94). Mean pre-procedure NIHSS score was 20.4. The MCA (36 %) and the posterior circulation (38 %) were the most frequently occluded areas. The clot was accessible in 85 % of the patients. Hemorrhage occurred in 22 % of the patients. Of 81 patients with concurrent thrombolysis, 18.5 % had hemorrhage compared with 27.3 % of 66 patients without thrombolysis ($p = 0.21$). Of the 126 patients with accessible clots, 36 % had a good mRS (less than or equal to 2) and 29 % died; in patients with inaccessible clots, 24 % had a good mRS and 38 % died. Factors associated with clinical success were younger age ($p = 0.001$) and lower NIHSS score at admission to the hospital ($p = 0.001$). Compared with a matched cohort, patients who received mechanical intervention were 14.8 times more likely to have a good mRS (95 % CI: 4.4 to 50.0; $p < 0.001$). The authors concluded that percutaneous mechanical embolectomy in the treatment of AIS is feasible and seems to provide an option for some patients seen after the interval for administration of IV tPA therapy has elapsed.

Broderick (2009) reviewed advances in ET for AIS. Data from primate studies, randomized studies of IV r-tPA, as well as non-randomized and randomized studies of ET were reviewed. Clinical trial data demonstrate the superiority of endovascular treatment with thrombolytic medication or mechanical methods to re-open arteries compared with control patients from the PROACT II Trial treated with heparin alone. However, these same clinical trials, as well as pre-clinical primate models, indicate that re-canalization, whether by endovascular approaches or standard-dose r-tPA, is unlikely to improve clinical outcome after a certain time point. Although the threshold beyond which re-perfusion has no or little benefit has yet to be conclusively defined, accumulated data to this point indicate an overall threshold of approximately 6 to 7 hours. In addition, although the risk of symptomatic intra-cerebral hemorrhage is similar in trials of IV lytics and endovascular approaches, endovascular approaches have distinctive risk profiles that can impact outcome. The author concluded that the treatment of AIS is evolving with new tools to re-open arteries and salvage the ischemic brain. Ongoing randomized trials of these new approaches are prerequisite next steps to demonstrate whether re-perfusion translates into clinical effectiveness.

The American Heart Association/American Stroke Association Stroke Council/Clinical Cardiology Council's guidelines for the early management of adults with ischemic stroke (Adams et al, 2007) stated that:

- Although the Merci device is a reasonable intervention for extraction of IA thrombi in carefully selected patients, the panel also recognizes that the utility of the device in improving outcomes after stroke is unclear. The panel also recommends that the device be studied in additional clinical trials that will define its role in the emergency management of stroke.
- The usefulness of other mechanical endovascular treatments is not established. These devices should be used in the setting of clinical trials.

Guidelines from the Institute for Clinical Systems Improvement (ICSI, 2010) state that the utility of mechanical embolectomy devices, including the MERCI and Penumbra, in improving clinical outcomes "remains unclear." Furthermore, the Australian National Stroke Foundation's clinical guidelines for acute stroke management (2010) found insufficient evidence for the use of mechanical embolectomy. The Stroke Foundation of

New Zealand (2010) reached similar conclusions about the lack of evidence for mechanical embolectomy. Guidelines on stroke from the Scottish Intercollegiate Guidelines Network (SIGN, 2008) concluded: "Mechanical clot retrieval devices should be further evaluated in randomised controlled trials." The National Institute for Health and Clinical Excellence's clinical guideline on the diagnosis and initial management of acute stroke and transient ischemic attack (NICE, 2008) did not mention the use of mechanical embolectomy.

In addition, the American Heart Association's scientific statement on indications for the performance of intra-cranial endovascular neuro-interventional procedures (Meyers et al, 2009) stated that;

The Trevo Retriever (Concentric Medical, Mountain View, CA) is designed to remove thrombus in patients with acute stroke. There is currently a clinical trial that evaluates the performance of the Trevo Retriever versus the Merci Retriever in restoring blood flow to the brain of patients experiencing an acute ischemic stroke in a large vessel (the TREV02 Trial).

Baker and colleagues (2011) described the state of the evidence supporting use of neuro-thrombectomy devices in the treatment of AIS. Medline, SCOPUS, the Cochrane Central Register of Controlled Trials, the Cochrane Database of Systematic Reviews, and Web of Science were searched, without language restrictions, from their inception through May 2010. The Medline and Cochrane Central Register of Controlled Trials searches were updated through November 2010. Two independent investigators screened citations for human studies of any design or case series or case reports of patients with an AIS that evaluated a neuro-thrombectomy device and reported at least 1 clinical effectiveness outcome or harm. Using standardized protocols, 2 independent investigators extracted information about study characteristics and outcomes, and a third reviewer resolved disagreement. A total of 87 articles met eligibility criteria, including 18 prospective single-group studies, 7 non-comparative retrospective studies, and 62 case series or case reports. Two FDA-cleared devices, the MERCI Retriever (Concentric Medical, Mountain View, CA) (40 %) and the Penumbra System (Penumbra, Alameda, CA) (9 %), represented a large portion of the available data. All prospective and retrospective studies provided data on successful recanalization with widely varying rates (43 % to 78 % with the MERCI Retriever and 83 % to 100 % with the Penumbra System). Rates of harms, including symptomatic (16 studies; 0 % to 10 % with the MERCI Retriever and 0 % to 11 % with the Penumbra System) or asymptomatic (13 studies; 28 % to 43 % and 1 % to 30 %, respectively) intra-cranial hemorrhage and vessel perforation or dissection (11 studies; 0 % to 7 % and 0 % to 5 %, respectively), also varied by device. Predictors of harm included older age, history of stroke, and higher baseline stroke severity scores, whereas successful recanalization was the sole predictor of good outcomes. Most available data are from single-group, non-comparative studies. In addition, the patient population most likely to benefit from these devices is undetermined. The authors concluded that currently available neuro-thrombectomy devices offer intriguing treatment options in patients with AIS. They stated that future trials should use a randomized design, with adequate power to show equivalency or non-inferiority between competing strategies or devices, and strive to identify populations that are most likely to benefit from use of neuro-thrombectomy devices. Furthermore, future studies should also examine if neuro-thrombectomy devices affect final health outcomes associated with stroke rather than improving re-canalization alone when compared with contemporary controls.

In an editorial that accompanied the afore-mentioned study, Khatri (2011) stated that "[t]he uncertainty about the effectiveness of neuro-thrombectomy is unsettling ... [i]f neuro-thrombectomy devices do not provide net benefit beyond that of intravenous rt-PA, then our current use of these devices is wasteful and perhaps even harmful ... It is premature to compare different devices until we first establish the clinical benefit of neuro-thrombectomy over thrombolysis. Until then, expansion of the use of neuro-thrombectomy is unjustified". Furthermore, in a review on endovascular stroke treatment, Grunwald et al (2011) noted that mechanical re-canalization devices can potentially have a clinically relevant impact in the interventional treatment of stroke, but at the present time, a randomized study would be beneficial.

Noting that earlier trials of endovascular therapy for ischemic stroke have produced variable results, Campbell, et al. (2015) conducted a study to test whether more advanced imaging selection, recently developed devices, and earlier intervention improve outcomes. The investigators randomly assigned patients with ischemic stroke who were receiving 0.9 mg of alteplase per kilogram of body weight less than 4.5 hours after the onset of

ischemic stroke either to undergo endovascular thrombectomy with the Solitaire FR (Flow Restoration) stent retriever or to continue receiving alteplase alone. All the patients had occlusion of the internal carotid or middle cerebral artery and evidence of salvageable brain tissue and ischemic core of less than 70 ml on computed tomographic (CT) perfusion imaging. The coprimary outcomes were reperfusion at 24 hours and early neurologic improvement (≥ 8 -point reduction on the National Institutes of Health Stroke Scale or a score of 0 or 1 at day 3). Secondary outcomes included the functional score on the modified Rankin scale at 90 days. Results The trial was stopped early because of efficacy after 70 patients had undergone randomization (35 patients in each group). The percentage of ischemic territory that had undergone reperfusion at 24 hours was greater in the endovascular-therapy group than in the alteplase-only group (median, 100% vs. 37%; $P < 0.001$). Endovascular therapy, initiated at a median of 210 minutes after the onset of stroke, increased early neurologic improvement at 3 days (80% vs. 37%, $P = 0.002$) and improved the functional outcome at 90 days, with more patients achieving functional independence (score of 0 to 2 on the modified Rankin scale, 71% vs. 40%; $P = 0.01$). There were no significant differences in rates of death or symptomatic intracerebral hemorrhage. The investigators concluded that, in patients with ischemic stroke with a proximal cerebral arterial occlusion and salvageable tissue on CT perfusion imaging, early thrombectomy with the Solitaire FR stent retriever, as compared with alteplase alone, improved reperfusion, early neurologic recovery, and functional outcome.

Goyal et al (2015) evaluated rapid endovascular treatment in addition to standard care in patients with acute ischemic stroke with a small infarct core, a proximal intracranial arterial occlusion, and moderate-to-good collateral circulation. The investigators randomly assigned participants to receive standard care (control group) or standard care plus endovascular treatment with the use of available thrombectomy devices (intervention group). Patients with a proximal intracranial occlusion in the anterior circulation were included up to 12 hours after symptom onset. Patients with a large infarct core or poor collateral circulation on computed tomography (CT) and CT angiography were excluded. Workflow times were measured against predetermined targets. The primary outcome was the score on the modified Rankin scale (range of 0 [no symptoms] to 6 [death]) at 90 days. A proportional odds model was used to calculate the common odds ratio as a measure of the likelihood that the intervention would lead to lower scores on the modified Rankin scale than would control care (shift analysis). The trial was stopped early because of efficacy. At 22 centers worldwide, 316 participants were enrolled, of whom 238 received intravenous alteplase (120 in the intervention group and 118 in the control group). In the intervention group, the median time from study CT of the head to first reperfusion was 84 minutes. The rate of functional independence (90-day modified Rankin score of 0 to 2) was increased with the intervention (53.0 %, versus 29.3 % in the control group; $p < 0.001$). The primary outcome favored the intervention (common odds ratio [OR], 2.6; 95 % CI: 1.7 to 3.8; $p < 0.001$), and the intervention was associated with reduced mortality (10.4 %, versus 19.0 % in the control group; $p = 0.04$). Symptomatic intracerebral hemorrhage occurred in 3.6 % of participants in intervention group and 2.7 % of participants in control group ($p = 0.75$). The investigators concluded that, among patients with acute ischemic stroke with a proximal vessel occlusion, a small infarct core, and moderate-to-good collateral circulation, rapid endovascular treatment improved functional outcomes and reduced mortality.

Guidelines from the American Heart Association and the American Stroke Association (Powers et al, 2015) systematically review the clinical trials of endovascular therapy for stroke. The guidelines recommend that patients should receive endovascular therapy with a stent retriever if they meet all the following criteria (Class I; Level of Evidence A):

1. Pre-stroke mRS score 0 to 1,
2. Acute ischemic stroke receiving intravenous r-tPA within 4.5 hours of onset according to guidelines from professional medical societies,
3. Causative occlusion of the internal carotid artery or proximal MCA (M1),
4. Age greater than or equal to 18 years,
5. NIHSS score of greater than or equal to 6,
6. ASPECTS of greater than or equal to 6, and
7. Treatment can be initiated (groin puncture) within 6 hours of symptom onset.

Based on lower levels of evidence, the guidelines recommend that, in carefully selected patients with anterior circulation occlusion who have contraindications to intravenous r-tPA, endovascular therapy with stent retrievers completed within 6 hours of stroke onset is reasonable. The guidelines also state that, although the benefits are uncertain, use of endovascular therapy with stent retrievers may be reasonable for carefully selected patients with acute ischemic stroke in whom treatment can be initiated (groin puncture) within 6 hours of symptom onset and who have causative occlusion of the M2 or M3 portion of the MCAs, anterior cerebral arteries, vertebral arteries, basilar artery, or posterior cerebral arteries.

The guidelines (Powers et al, 2015) state that the usefulness of mechanical thrombectomy devices other than stent retrievers is not well established, either for technical efficacy or clinical benefit. The guidelines explain that most of the patients in the pivotal studies who underwent an endovascular procedure were treated with a stent retriever. These trials were not designed to demonstrate the superiority of stent retrievers over other devices, such as snares or suction aspiration systems. The guidelines note that there are no published randomized clinical trials demonstrating clinical benefit nor comparing its relative effectiveness of other devices versus stent retrievers.

In summary, although earlier studies of mechanical embolectomy have shown mixed results, newer studies have shown improved outcomes in patients with acute ischemic stroke of the anterior circulation who are selected using advanced imaging and treated using newer stent retrievers.

Cerebral vasospasm remains a major source of morbidity and death in patients with aneurysmal SAH. When vasospasm becomes refractory to maximal medical management consisting of induced hypertension and hypervolemia and administration of calcium channel antagonists, endovascular therapies should be considered. The primary goal of endovascular treatment is to increase cerebral blood flow (CBF) to prevent cerebral infarction. Two of the more frequently studied endovascular treatments are transluminal balloon angioplasty (TBA) and IA papaverine infusion. Other pharmacological vasodilating agents for the treatment of cerebral vasospasm include IA nimodipine, nicardipine, verapamil, and milrinone.

Elliott and colleagues (1998) tested the hypothesis that TBA is superior to papaverine infusion for the treatment of proximal anterior circulation arterial vasospasm following SAH. Between 1989 and 1995, 125 vasospastic distal internal carotid artery or proximal MCA vessel segments were treated in 52 patients. Blood flow velocities of the involved vessels were assessed by using transcranial Doppler (TCD) monitoring in relation to the day of treatment with TBA or papaverine infusion. Balloon angioplasty and papaverine infusion cohorts were compared based on mean pre- and post-treatment velocity at 24 and 48 hrs using the 1-tailed, paired-samples t-test. Balloon angioplasty alone was performed in 101 vessel segments (81 %) in 39 patients (75 %), whereas papaverine infusion alone was used in 24 vessel segments (19 %) in 13 patients (25 %). Although repeated treatment after TBA was needed in only 1 vessel segment, repeated treatment following papaverine infusion was required in 10 vessel segments (42 %) in 6 patients because of recurrent vasospasm ($p < 0.001$). Seven vessel segments (29 %) with recurrent spasm after papaverine infusion were treated with TBA. Although vessel segments treated with papaverine demonstrated a 20 % mean decrease in blood flow velocity ($p < 0.009$) on post-treatment Day 1, velocities were not significantly lower than pre-treatment levels by post-treatment Day 2 ($p = 0.133$). Balloon angioplasty resulted in a 45 % mean decrease in velocity to a normal level following treatment ($p < 0.001$), a decrease that was sustained. The authors concluded that TBA is superior to papaverine infusion for the permanent treatment of proximal anterior circulation vasospasm following aneurysmal SAH.

Hoh and Ogilvy (2005) reviewed clinical series of endovascular treatment of cerebral vasospasm reported in the English language literature. Transluminal balloon angioplasty produced clinical improvement in 62 % of patients, significantly improved mean TCD velocities ($p < 0.05$), significantly improved CBF in 85 % of patients as studied by (133)Xenon techniques and serial single photon emission computerized tomography, and was associated with 5.0 % complications and 1.1 % vessel rupture. Intra-arterial papaverine therapy produced clinical improvement in 43 % of patients but only transiently, requiring multiple treatment sessions (1.7 treatments per patient); significantly improved mean TCD velocities ($p < 0.01$) but only for less than 48 hours; improved CBF in 60 % of patients but only for less than 12 hours; and was associated with increases in intra-cranial pressure and 9.9 % complications. Intra-arterial nicardipine therapy produced clinical improvement in 42 % of patients, significantly improved mean TCD velocities ($p < 0.001$)

for 4 days, and was associated with no complications in the authors' small series. These investigators have adopted a treatment protocol at their institution of TBA and IA nicardipine therapy as the endovascular treatments for medically refractory cerebral vasospasm.

In a review on neuro-interventional for the treatment of vasospasm, Brisman and colleagues (2006) stated that they favor the use of TBA over IA anti-spasmodics due to the increased durability and long-lasting effects of the former and lower risk profile. Also, in a review on endovascular treatment of vasospasm following SAH, Abdennour and associates (2007) noted that in Europe, nimodipine is widely used whereas nicardipine and verapamil are the major molecules administered in North America where intravenous nimodipine is not FDA-approved. Papaverine is less used nowadays because of its short duration of action and of the risk of aggravation of raised intra-cranial pressure.

Furthermore, Platz and associates (2008) reported a possible new side effect of IA administration of papaverine. After the treatment of cerebral vasospasm in a SAH patient by IA papaverine into the left posterior cerebral artery, severe mesencephalic extravasation of blood and contrast media was detected. After reviewing the literature, the authors concluded that interruption of the blood-brain barrier by papaverine most likely combined with a secondary hyper-perfusion phenomena, and perhaps a direct toxic effect on brain tissue was the mechanism of this major complication. They stated that in treating vasospasm in areas with a high density of perforating arteries, especially in the posterior circulation, papaverine should be used cautiously because a safe regimen has yet to be established. In this situation, alternative agents such as calcium channel blockers could be considered, but evidence-based data are still missing.

Guidelines from the American Academy of Neurology on subarachnoid hemorrhage (Connolly et al, 2012) stated: "Cerebral angioplasty and/or selective intra-arterial vasodilator therapy is reasonable in patients with symptomatic cerebral vasospasm, particularly those who are not rapidly responding to hypertensive therapy (Class IIa; Level of Evidence B)". The guidelines explain: Endovascular intervention is often used in patients who do not improve with hemodynamic augmentation and those with sudden focal neurological deficits and focal lesions on angiography referable to their symptoms. Interventions generally consist of balloon angioplasty for accessible lesions and vasodilator infusion for more distal vessels. Many different vasodilators are in use. In general, these are calcium channel blockers, but nitric oxide donors have been used in small series as well. Papaverine is used less frequently because it can produce neurotoxicity. The primary limitation of vasodilator therapy is the short duration of benefit. As with hemodynamic augmentation, there have been no randomized trials of these interventions, but large case series have demonstrated angiographic and clinical improvement." The guidelines recommend against stenting, stating that "Stenting of a ruptured aneurysm is associated with increased morbidity and mortality, and should only be considered when less risky options have been excluded (Class III; Level of Evidence C)."

Guidelines from the Neurocritical Care Society (Diringer et al, 2011) state: "Endovascular treatment using intra-arterial vasodilators and/or angioplasty may be considered for vasospasm-related DCI [delayed cerebral ischemia] (moderate quality evidence-strong recommendation)." The guidelines state that "the timing and triggers of endovascular treatment of vasospasm remains unclear, but generally rescue therapy for ischemic symptoms that remain refractory to medical treatment should be considered. The exact timing is a complex decision which should consider the aggressiveness of the hemodynamic intervention, the patients' ability to tolerate it, prior evidence of large artery narrowing, and the availability of and the willingness to perform angioplasty or infusion of intra-arterial agents (moderate quality evidence-strong recommendation)." The guidelines, however, recommend against prophylactic endovascular treatment. "The use of routine prophylactic cerebral angioplasty is not recommended (High quality Evidence-Strong Recommendation)." The guidelines explain: "Most studies are retrospective case series or comparison studies, with few prospective studies. Hence, the literature has demonstrated the feasibility, durability, and safety profile of intra-arterial vasodilator therapy and angioplasty, and the combination of the two, but has not demonstrated this for newer methods. The literature has not provided sufficient information regarding timing of the endovascular rescue therapy nor the optimum number of repeat treatments necessary. However, the single randomized controlled trial of prophylactic angioplasty, done early after SAH without the presence of angiographic arterial narrowing, suggested a lower risk of DCI, albeit at a risk of vessel rupture and death from the procedure and ultimately no difference in outcome [citing Zwienenberg-Lee, et al., 2008]. There are presently insufficient data to determine if intraarterial

vasodilator therapy alone, or angioplasty alone, or a combination of treatments is superior to one another or superior to medical treatment alone."

By contrast, international guidelines from the European Stroke Organization on management of intracranial aneurysms and subarachnoid hemorrhage (Steiner, et al., 2013) have no recommendations for angioplasty or intra-arterial vasodilators. In the Canadian best practice recommendations for management of SAH and intracerebral hemorrhage (Canadian Stroke Network/Heart & Stroke Foundation of Canada, 2006), IA papaverine was not mentioned as a option.

In a Cochrane review, Rinkel et al (2005) examined if calcium antagonists improve outcome in patients with aneurysmal SAH. These investigators searched the Cochrane Stroke Group Trials Register (September 2003). In addition, they searched Medline (1966 to October 2003) and EMBASE (1980 to October 2003), hand-searched 2 Russian journals (1990 to 2003) and contacted trialists and pharmaceutical companies (in 1995 and 1996) to identify further studies. All unconfounded, truly randomized controlled trials comparing any calcium antagonist with control were included in this analysis. Two reviewers independently extracted the data and assessed trial quality. Trialists were contacted to obtain missing information. These researchers analyzed 12 trials totaling 2,844 patients with SAH (1,396 in the treatment group and 1,448 in the control group). The drugs analyzed were: nimodipine (8 trials, 1,574 patients), nicardipine (2 trials, 954 patients), AT877 (1 trial, 276 patients) and magnesium (1 trial, 40 patients). Overall, calcium antagonists reduced the risk of poor outcome: RR 0.82 (95 % CI: 0.72 to 0.93); the absolute risk reduction was 5.1 %, the corresponding number of patients needed to treat to prevent a single poor outcome event was 20. For oral nimodipine alone the RR was 0.70 (0.58 to 0.84). The RR of death on treatment with calcium antagonists was 0.90 (95 % CI: 0.76 to 1.07), that of clinical signs of secondary ischemia 0.67 (95 % CI: 0.60 to 0.76), and that of CT- or MR-confirmed infarction 0.80 (95 % CI: 0.71 to 0.89). The authors concluded that calcium antagonists reduce the risk of poor outcome and secondary ischemia after aneurysmal SAH. The results for "poor outcome" depend largely on a single large trial with oral nimodipine; the evidence for nicardipine, AT877 and magnesium is inconclusive. The evidence for nimodipine is not beyond every doubt, but given the potential benefits and modest risks of this treatment, against the background of a devastating natural history, oral nimodipine (60 mg every 4 hours) is currently indicated in patients with aneurysmal SAH. Intravenous administration of calcium antagonists can not be recommended for routine practice on the basis of the present evidence.

Weyer et al (2006) noted that cerebral vasospasm and delayed cerebral ischemia remain common complications of aneurysmal SAH, and yet therapies for cerebral vasospasm are limited. Despite a large number of clinical trials, only calcium antagonists have strong evidence supporting their effectiveness. These investigators performed a systematic review of the literature on the treatment of cerebral vasospasm. A literature search for randomized controlled trials of therapies used for prevention or treatment of cerebral vasospasm and/or delayed cerebral ischemia was conducted, and 41 articles meeting the review criteria were found. Study characteristics and primary results of these articles are reviewed. Key indicators of quality were poor when averaged across all studies, but have improved greatly over time. The only proven therapy for vasospasm is nimodipine. Tirilazad is not effective, and studies of hemodynamic maneuvers, magnesium, statin medications, endothelin antagonists, steroid drugs, anti-coagulant/anti-platelet agents, and intra-theal fibrinolytic drugs have yielded inconclusive results. The following conclusions were made: nimodipine is indicated after SAH and tirilazad is not effective. More study of hemodynamic maneuvers, the effectiveness of other calcium channel antagonists such as nicardipine delivered by other routes (e.g., intrathecally), magnesium, statin drugs, endothelin antagonists, and intra-theal fibrinolytic therapy is warranted.

Shah et al (2009) examined the safety and tolerability of super-selective intra-arterial magnesium sulfate in combination with intra-arterial nicardipine in patients with cerebral vasospasm after SAH. Patients were treated in a prospective protocol at 2 teaching medical centers. Emergent cerebral angiography was performed if there was either clinical, ultrasound, and/or CT perfusion deficits suggestive of cerebral vasospasm. Intra-arterial magnesium sulfate (0.25 to 1 g) was administered via a microcatheter in the affected vessels in combination with nicardipine (2.5 to 20.0 mg). Mean arterial pressures (MAP) and intra-cranial pressures (ICP) were monitored during the infusion. Immediate and sustained angiographic and clinical improvement was determined from post-treatment angiograms and clinical follow-up. Angiographical and clinical outcomes were compared to 2 published case series that has used nicardipine alone. A

total of 58 vessels were treated in 14 patients (mean age of 42 years; 11 women) with acute SAH. The treatment was either intra-arterial nicardipine and magnesium sulfate alone or in conjunction with primary angioplasty. Forty vessels (69 %) had immediate angiographical improvement with intra-arterial nicardipine and magnesium sulfate alone and 18 vessels (31 %) required concomitant balloon angioplasty with complete reversal of the vasospasm. Re-treatment was required in 13 vessels (22 %) and the median time for retreatment was 2 days (range of 1 to 13 days). Nicardipine treatment resulted in the reduction of MAP (12.3 mmHg, standard error [SE] 1.34, $p < 0.0001$) without any significant change in ICP. Magnesium sulfate infusion was not associated with change in MAP or ICP. Among 31 procedures, immediate neurological improvement was observed in 22 (71 %) procedures. In 12 (86 %) patients, there were no infarctions in the follow-up CT scan acquired between 24 and 48 hrs. No statistical significant difference was observed in angiographical and clinical outcome of patients treated with the combination therapy in comparison with historical controls treated with nicardipine alone. The authors concluded that administration of intra-arterial magnesium sulfate in combination with nicardipine was well-tolerated in patients with SAH and cerebral vasospasm without a significant change in MAP and ICP. They stated that the efficacy of this combination therapy should be evaluated in a larger, controlled setting.

Reddy and Yeh (2009) stated that injectable nicardipine is increasingly being used to manage neurovascular conditions. To better understand its place in therapy, these investigators conducted an evidenced-based literature review. A total of 223 article abstracts were identified; after independent review by 2 individuals and a supplemental manual search, 29 were deemed relevant and were included in this review. Nicardipine has been studied or recommended for management of hypertension in many neurovascular settings (e.g., ischemic stroke, intra-cerebral hemorrhage, craniotomy, and spinal surgery), for vasospasm in aneurysmal SAH, and in acute traumatic brain injury. In the management of hypertension in acute stroke, nicardipine is one of several recommended options available; expert opinion forms the basis of these recommendations in clinical guidelines, with limited randomized controlled trial evidence to support its use. Among the various anti-hypertensive agents, nicardipine has the highest drug acquisition cost.

In 2 meta-analyses, intravenous nicardipine had no impact on patient outcomes (death, disability) in patients with acute traumatic brain injury (RR 0.25, 95 % CI: 0.05 to 1.27) or in patients with aneurysmal SAH (RR 0.97, 95 % CI: 0.78 to 1.20). Intra-arterial nicardipine reduced angiographical diameter (p value not reported) and peak systolic velocities on transcranial Doppler images ($p < 0.001$) in published case series. Given nicardipine's high cost relative to that of other agents and the limited evidence to support its use in patients with neurovascular conditions, this drug should be considered only in patients who have failed or have contraindications to alternative agents in the management of hypertension. The authors stated that although intra-arterial nicardipine appears to be promising in aneurysmal SAH, well-designed studies are needed in this setting before its use can be routinely recommended.

In a Cochrane review, Zhang et al (2010) evaluated the safety and effectiveness of tirilazad (a non-glucocorticoid, 21-aminosteroid that inhibits lipid peroxidation) in patients with aneurysmal SAH. These investigators searched the Cochrane Stroke Group Trials Register (last searched October 2009); the Cochrane Central Register of Controlled Trials (CENTRAL) (The Cochrane Library Issue 2, 2009); MEDLINE (1966 to October 2009); EMBASE (1980 to October 2009); and the Stroke Trials Directory, the National Center for Complementary and Alternative Medicine, and the National Institute of Health Clinical Trials Database (searched October 2009). They hand-searched 10 Chinese journals, searched the reference lists of relevant publications, and contacted the manufacturers of tirilazad. Randomised trials of tirilazad started within 4 days of SAH onset, compared with placebo or open control in patients with aneurysmal SAH documented by angiography and computerised tomography scan or cerebrospinal fluid examination, or both. These researchers extracted data relating to case fatality, poor outcome (death, vegetative state, or severe disability), delayed cerebral ischemia (or symptomatic vasospasm), cerebral infarction and adverse events of treatments. They pooled the data using the Peto fixed-effect method for dichotomous data. The authors included 5 double-blind, placebo-controlled trials involving 3,821 patients; there was no significant heterogeneity. Oral or intravenous nimodipine was used routinely as a background treatment in both groups in all trials. There was no significant difference between the 2 groups at the end of follow-up for the primary outcome, death (OR 0.89, 95 % CI: 0.74 to 1.06), or in poor outcome (death, vegetative state or severe disability) (OR 1.04, 95 % CI: 0.90 to 1.21). During the treatment period, fewer patients developed delayed cerebral ischemia in the tirilazad group than in the control group (OR 0.80, 95 % CI: 0.69 to 0.93). Subgroup analyses did not demonstrate any significant difference in

effects of tirilazad on clinical outcomes. Leukocytosis and prolongation of Q-T interval occurred significantly more frequently in the treatment group in only 1 trial evaluating tirilazad at high dose. There was no significant difference in infusion site disorders or other laboratory parameters between the 2 groups. The authors concluded that there is no evidence that tirilazad, in addition to nimodipine, reduces mortality or improves poor outcome in patients with aneurysmal SAH.

Tenser and colleagues (2011) reviewed the initial studies of the Merci Retriever and Penumbra System for mechanical clot extraction. Baseline patient characteristics, as well as revascularization rates and clinical outcome, were examined. Baseline National Institutes of Health Stroke Scale scores were greater than those observed in previous IV tPA studies, consistent with large-vessel occlusion. Successful re-canalization occurred more frequently than with IV tPA and was associated with improved clinical outcome and mortality. Symptomatic intra-cranial hemorrhage and mortality rates were greater than those seen with IV tPA. Mechanical clot extraction can be performed safely in patients with large-vessel occlusions, and successful re-canalization resulted in better clinical outcomes than those without. The authors concluded that mechanical thrombectomy provides a therapeutic option for ischemic stroke patients who are ineligible for, or who do not respond to, IV thrombolytics. Moreover, they stated that further studies, including randomized clinical trials, are needed to validate these findings.

In a review on "Mechanical thrombectomy devices for treatment of stroke" Radoslav and Saver (2012) stated that "Mechanical thrombectomy devices work well in large, proximal arteries, rapidly debulking large clot burdens that are resistant to chemical fibrinolysis. Conversely, mechanical thrombectomy devices are currently not well suited for distal arterial branches (hard to navigate to and device diameters too large) and are not options for penetrator occlusions, while chemical fibrinolysis works well on those targets Definitive data regarding the efficacy of mechanical thrombectomy devices in improving final clinical outcome over medical therapy alone awaits the conclusion of ongoing trials. One trial, the Interventional Management of Stroke 3 (IMS 3) Trial, has been stopped for futility. However, the implications for IMS 3 for mechanical thrombectomy may be limited, as many patients enrolled in the endovascular arm of IMS 3 were treated with intra-arterial fibrinolytics drugs rather than mechanical thrombectomy, essentially none of the patients were treated with the most technically efficacious device class, the stent retrievers, and the trial included large number of patients with no occlusions or small, distal occlusions, which are less likely to benefit from mechanical retrieval".

Furthermore, an UpToDate review on "Reperfusion therapy for acute ischemic stroke" (Oliveira-Filho and Samuels, 2012) states that "The 2012 ACCP guidelines concluded that the available data regarding mechanical thrombectomy in acute ischemic stroke are of low quality and leave considerable uncertainty regarding the impact of this intervention on survival and functional outcome. A 2011 systematic review identified 87 studies published through November 2010 that evaluated mechanical clot disruption devices for acute ischemic stroke. All 87 studies were uncontrolled, only 18 were prospective, and only 3 of those used blinded outcome assessment. Nearly half of the data came from studies evaluating the Merci Retriever and the Penumbra System (40 % and 9 %, respectively). For these and other thrombectomy devices, the included studies reported widely varying outcome rates of successful recanalization, good clinical outcome, symptomatic intracranial hemorrhage, vessel injury, and mortality. The one clear finding was that successful recanalization predicted a good outcome. However, given the methodologic limitations of the available studies, the investigators concluded that randomized trials are necessary to establish whether mechanical clot disruption devices improve patient outcome Thus, although the Merci and Penumbra devices are approved for clot removal in carefully selected patients, their clinical utility for improving outcomes after stroke is unproven. Further study in randomized controlled trials is needed before the role of mechanical clot disruption devices is defined for the emergency management of acute ischemic stroke".

Alshekhlee et al (2012) stated that mechanical thrombectomy is a promising adjuvant or stand-alone therapy for AIS caused by occlusion of a large vessel in patients beyond the systemic thrombolysis therapeutic window. These investigators reviewed the clinical and angiographical outcomes of mechanical thrombectomy with use of the Merci retriever device. Available literature published to date on the major trials and observational studies involving the Merci retriever was reviewed. In addition to the review, results from studies involving the Merci retriever were compared to results from Prolyse in Acute Cerebral Thromboembolism II (PROACT II) and the Penumbra device studies. The predictors

for favorable outcome following re-vascularization with the Merci device were reviewed on the basis of published stratified analyses. Favorable clinical outcome was defined in the Merci experience by a mRS score of less than or equal to 2 at 90 days following AIS. Presented in this review were a total of 1,226 patients treated with the Merci device; 305 patients were from 2 pivotal trials involving the device, and the remaining 921 patients were from observational studies in the Merci registry. The 90-day mRS score of less than or equal to 2 was achieved in 32 % of the patient group, with an overall mortality rate of 35.2 %. Symptomatic intra-cerebral hemorrhage was identified in 7.3 % of patients treated with Merci retriever, a result comparable to that in the PROACT II and Penumbra thrombectomy trials. Successful re-canalization, lower NIHSS score, and younger age were identified as the strongest predictors of favorable outcomes. The authors concluded that mechanical thrombectomy with the Merci retriever device is a safe treatment modality for AIS patients presenting with a large-vessel occlusion within 8 hours of symptom onset. Moreover, they stated that although the Merci retriever showed a good re-canalization rate, there are currently no randomized clinical trials to assess its clinical effectiveness in comparison with systemic thrombolysis within a window of 3 to 4.5 hours or with standard of care beyond a 4.5-hour window.

Hussain et al (2012) noted that the effectiveness of IV systemic thrombolysis is limited in patients with severe AIS and large-vessel occlusion. Mechanical thrombectomy has been the mainstay therapy in large-vessel occlusion. These investigators reviewed the evidence regarding the Penumbra aspiration device. The pivotal single-arm prospective trial that led to its approval by the FDA enrolled 125 patients within 8 hours of symptom onset and demonstrated an 82 % re-canalization rate, to TIMI scores of 2 and 3. The risk of symptomatic intra-cranial hemorrhage was 10 %, and a mRS score of less than or equal to 2 was 25 %. In the post-marketing registry, 157 vessels were treated, with 87 % achieving TIMI 2 and 3 re-canalization and 41 % having a mRS score of less than or equal to 2. The authors concluded that the Penumbra aspiration system is an effective tool to safely re-vascularize large-vessel occlusions in patients within 8 hours of onset of AIS who are either refractory to or excluded from IV thrombolytic therapy. Moreover, they stated that further prospective, randomized controlled trials will be needed to address whether this ability translates into neurologic improvement and better functional outcomes.

San Roman et al (2012) examined the safety and effectiveness of the new TREVO stent-like retriever in consecutive patients with acute stroke. These researchers conducted a prospective, single-center study of 60 patients (mean age of 71.3 years; male 47 %) with stroke lasting less than 8 hours in the anterior circulation (n = 54) or less than 12 hours in the vertebra-basilar circulation (n = 6) treated if CT perfusion/CT angiography confirmed a large artery occlusion, ruled out a malignant profile, or showed target mismatch if symptoms were greater than 4.5 hours. Successful re-canalization (Thrombolysis In Cerebral Infarction 2b-3), good outcome (a mRS score of 0 to 2) and mortality at Day 90, device-related complications, and symptomatic hemorrhage (parenchymal hematoma Type 1 or parenchymal hematoma Type 2 and NIHSS score increment greater than or equal to 4 points) were prospectively assessed. Median (interquartile range) NIHSS score on admission was 18 (12 to 22). The median (interquartile range) time from stroke onset to groin puncture was 210 (173 to 296) minutes. Successful re-vascularization was obtained in 44 (73.3 %) of the cases when only the TREVO device was used and in 52 (86.7 %) when other devices or additional intra-arterial t-PA were also required. The median time (interquartile range) of the procedure was 80 (45 to 114) minutes. Good outcome was achieved in 27 (45 %) of the patients and the mortality rate was 28.3 %. Seven patients (11.7 %) presented a symptomatic intra-cranial hemorrhage. No other major complications were detected. The authors concluded that the TREVO device was reasonably safe and effective in patients with severe stroke. They stated that these results support further investigation of the TREVO device in multi-centric registries and randomized clinical trials.

Broderick et al (2013) stated that endovascular therapy is increasingly used after the administration of intravenous tissue plasminogen activator (t-PA) for patients with moderate-to-severe AIS. These investigators examined if a combined approach is more effective than intravenous t-PA alone. They randomly assigned eligible patients who had received intravenous t-PA within 3 hours after symptom onset to receive additional endovascular therapy or intravenous t-PA alone, in a 2:1 ratio. The primary outcome measure was a mRS score of 2 or less (indicating functional independence) at 90 days (scores range from 0 to 6, with higher scores indicating greater disability). The study was stopped early because of futility after 656 participants had undergone randomization (434 patients to endovascular therapy and 222 to intravenous t-PA alone). The proportion

of participants with a mRS score of 2 or less at 90 days did not differ significantly according to treatment (40.8 % with endovascular therapy and 38.7 % with intravenous t-PA; absolute adjusted difference, 1.5 percentage points; 95 % CI: -6.1 to 9.1, with adjustment for the NIHSS score [8 to 19, indicating moderately severe stroke, or greater than or equal to 20, indicating severe stroke]), nor were there significant differences for the predefined subgroups of patients with an NIHSS score of 20 or higher (6.8 percentage points; 95 % CI: -4.4 to 18.1) and those with a score of 19 or lower (-1.0 percentage point; 95 % CI: -10.8 to 8.8). Findings in the endovascular-therapy and intravenous t-PA groups were similar for mortality at 90 days (19.1 % and 21.6 %, respectively; $p = 0.52$) and the proportion of patients with symptomatic intra-cerebral hemorrhage within 30 hours after initiation of t-PA (6.2 % and 5.9 %, respectively; $p = 0.83$). The authors concluded that the trial showed similar safety outcomes and no significant difference in functional independence with endovascular therapy after intravenous t-PA, as compared with intravenous t-PA alone.

Ciccone et al (2013) noted that in patients with ischemic stroke, endovascular treatment results in a higher rate of re-canalization of the affected cerebral artery than systemic intravenous thrombolytic therapy. These researchers compared the clinical effectiveness of the 2 approaches. They randomly assigned 362 patients with AIS, within 4.5 hours after onset, to endovascular therapy (intra-arterial thrombolysis with recombinant t-PA, mechanical clot disruption or retrieval, or a combination of these approaches) or intravenous t-PA. Treatments were to be given as soon as possible after randomization. The primary outcome was survival free of disability (defined as a mRS score of 0 or 1 on a scale of 0 to 6, with 0 indicating no symptoms, 1 no clinically significant disability despite symptoms, and 6 death) at 3 months. A total of 181 patients were assigned to receive endovascular therapy, and 181 intravenous t-PA. The median time from stroke onset to the start of treatment was 3.75 hours for endovascular therapy and 2.75 hours for intravenous t-PA ($p < 0.001$). At 3 months, 55 patients in the endovascular-therapy group (30.4 %) and 63 in the intravenous t-PA group (34.8 %) were alive without disability (OR adjusted for age, sex, stroke severity, and atrial fibrillation status at baseline, 0.71; 95 % CI: 0.44 to 1.14; $p = 0.16$). Fatal or non-fatal symptomatic intra-cranial hemorrhage within 7 days occurred in 6 % of the patients in each group, and there were no significant differences between groups in the rates of other serious adverse events or the case fatality rate. The authors concluded that the results of this trial in patients with AIS indicated that endovascular therapy is not superior to standard treatment with intravenous t-PA.

In an editorial that accompanied the afore-mention studies, Chimowitz (2013) stated that "A decision by Medicare to place a moratorium on reimbursement for endovascular treatment of acute ischemic stroke outside of randomized trials would facilitate recruitment in these urgently needed trials. Once the new trials are completed, endovascular treatment will have been given ample opportunity to prove itself".

An assessment by the BlueCross BlueShield Association Technology Evaluation Center (BCBSA, 2015) concluded that endovascular treatments for acute ischemic stroke did not meet the TEC criteria. The assessment stated that, for acute ischemic stroke, randomized controlled trials (RCTs) have not demonstrated a health benefit -- usually defined as a reduction in disability at 90 days -- for endovascular therapy compared with IV tPA. The Solitaire and Trevo devices appear to produce better outcomes than the Merci® device; how the 2 newer devices compare with each other is not known. Trials are under way that will provide more information on the value of endovascular treatments and possibly on the patient groups for whom they may be most effective.

This CLOTBUST device encircles the head and directs ultrasonic sound to obstructed arteries to speed up the dissolving effect of clot-busting tPA drugs (Alberta Heritage Foundation, 2014). This device is currently under study. The CLOTBUST headframe, developed by Cerevast Therapeutics, automatically targets the clot-affected region without the need for a trained sonographer or vascular technician to perform the procedure. Clinical studies are examining whether the addition of the acoustic energy of ultrasound to conventional intravenous tPA therapy dissolves blood clots more completely and provides stroke patients with better long-term outcomes compared to IV tPA therapy alone.

In a multi-center, open-label, pilot study, Barreto and associates (2013) examined the effects of tPA plus the CLOTBUST-HF, a novel operator-independent ultrasound device, in patients with ischemic stroke caused by proximal intra-cranial occlusion. All patients received standard-dose IV tPA, and shortly after tPA bolus, the CLOTBUST-HF device

delivered 2-hour therapeutic exposure to 2-MHz pulsed-wave ultrasound. Primary outcome was occurrence of symptomatic intra-cerebral hemorrhage. All patients underwent pre-treatment and post-treatment transcranial Doppler ultrasound or CT angiography; NIHSS scores were collected at 2 hours and modified Rankin scale at 90 days. Summary characteristics of all 20 enrolled patients were 60 % men, mean age of 63 (SD = 14) years, and median NIHSS of 15. Sites of pre-treatment occlusion were as follows: 14 of 20 (70 %) middle cerebral artery, 3 of 20 (15 %) terminal internal carotid artery, and 3 of 20 (15 %) vertebral artery. The median (interquartile range) time to tPA at the beginning of sonothrombolysis was 22 (13.5 to 29.0) minutes. All patients tolerated the entire 2 hours of insonation, and none developed symptomatic intra-cerebral hemorrhage. No serious adverse events were related to the study device. Rates of 2-hour re-canalization were as follows: 8 of 20 (40 %; 95 % CI: 19 % to 64 %) complete and 2 of 20 (10 %; 95 % CI: 1 % to 32 %) partial. Middle cerebral artery occlusions demonstrated the greatest complete re-canalization rate: 8 of 14 (57 %; 95 % CI: 29 % to 82 %). At 90 days, 5 of 20 (25 %, 95 % CI: 7 % to 49 %) patients had a modified Rankin scale of 0 to 1. The authors concluded that sonothrombolysis using a novel, operator-independent device, in combination with systemic tPA, seems safe, and re-canalization rates warrant evaluation in a phase III efficacy trial.

An UpToDate review on “Initial assessment and management of acute stroke” (Oliveira-Filho and Koroshetz, 2014) does not mention transcranial ultrasound as a therapeutic option.

Hong and colleagues (2014) noted that therapeutic hypothermia improves outcomes in experimental stroke models, especially after ischemia-reperfusion injury. In a prospective cohort study at 2 stroke centers, these researchers investigated the clinical and radiological effects of therapeutic hypothermia in AIS patients after re-canalization. They enrolled patients with AIS in the anterior circulation with an initial NIHSS greater than or equal to 10 who had successful re-canalization (greater than or equal to thrombolysis in cerebral ischemia, 2b). Patients at center A underwent a mild hypothermia (34.5° C) protocol, which included mechanical ventilation, and 48-hour hypothermia and 48-hour re-warming. Patients at center B were treated according to the guidelines without hypothermia. Cerebral edema, hemorrhagic transformation, good outcome (3-month modified Rankin Scale, less than or equal to 2), mortality, and safety profiles were compared. Potential variables at baseline and during the therapy were analyzed to evaluate for independent predictors of good outcome. The hypothermia group (n = 39) had less cerebral edema (p = 0.001), hemorrhagic transformation (p = 0.016), and better outcome (p = 0.017) compared with the normothermia group (n = 36). Mortality, hemi-craniectomy rate, and medical complications were not statistically different. After adjustment for potential confounders, therapeutic hypothermia (odds ratio, 3.0; 95 % confidence interval, 1.0-8.9; P=0.047) and distal occlusion (OR, 7.3; 95 % CI: 1.3 to 40.3; p = 0.022) were the independent predictors for good outcome. Absence of cerebral edema (OR, 5.4; 95 % CI: 1.6 to 18.2; p = 0.006) and no medical complications (OR, 9.3; 95 % CI: 2.2 to 39.9; p = 0.003) were also independent predictors for good outcome during the therapy. The authors concluded that in patients with ischemic stroke, after successful re-canalization, therapeutic hypothermia may reduce risk of cerebral edema and hemorrhagic transformation, and lead to improved clinical outcomes.

An UpToDate review on “Initial assessment and management of acute stroke” (Oliveira-Filho and Koroshetz, 2014) states that “Induced hypothermia is not currently recommended for patients with ischemic stroke, outside of clinical trials. An NINDS-funded randomized trial (ICTuS2/3) evaluating the combination of hypothermia and thrombolysis versus thrombolysis alone is currently underway”.

Transdermal Glyceryl Trinitrate

On behalf of the ENOS Trial Investigators, Bath et al (2015) evaluated outcomes after stroke in patients given drugs to lower their blood pressure (BP). In this multi-center, partial-factorial trial, these researchers randomly assigned patients admitted to hospital with an acute ischemic or hemorrhagic stroke and raised systolic BP (systolic 140 to 220 mm Hg) to 7 days of transdermal glyceryl trinitrate (GTN; 5 mg/day), started within 48 hours of stroke onset, or to no GTN (control group). A subset of patients who were taking anti-hypertensive drugs before their stroke were also randomly assigned to continue or stop taking these drugs. The primary outcome was function, assessed with the modified Rankin Scale at 90 days by observers masked to treatment assignment. Between July 20, 2001, and October 14, 2013, a total of 4,011 patients were enrolled. Mean BP was 167 (SD 19) mm Hg/90 (13) mm Hg at baseline (median of 26 hours [16 to 37] after stroke onset), and was significantly reduced on day 1 in 2,000 patients allocated to GTN

compared with 2,011 controls (difference -7.0 [95 % CI: -8.5 to -5.6] mm Hg/-3.5 [-4.4 to -2.6] mm Hg; both $p < 0.0001$), and on day 7 in 1,053 patients allocated to continue anti-hypertensive drugs compared with 1,044 patients randomized to stop them (difference -9.5 [95 % CI: -11.8 to -7.2] mm Hg/-5.0 [-6.4 to -3.7] mm Hg; both $p < 0.0001$). Functional outcome at day 90 did not differ in either treatment comparison-the adjusted common OR for worse outcome with GTN versus no-GTN was 1.01 (95 % CI: 0.91 to 1.13; $p = 0.83$), and with continue versus stop anti-hypertensive drugs OR was 1.05 (0.90 to 1.22; $p = 0.55$). The authors concluded that in patients with acute stroke and high BP, transdermal GTN lowered BP and had acceptable safety; but did not improve functional outcome. These investigators showed no evidence to support continuing pre-stroke anti-hypertensive drugs in patients in the first few days after acute stroke.

Krishnan et al (2016) noted that the Efficacy of Nitric Oxide in Stroke (ENOS) trial found that transdermal GTN (a nitric oxide donor) lowered BP but did not improve functional outcome in patients with acute stroke. However, GTN was associated with improved outcome if patients were randomized within 6 hours of stroke onset. In this pre-specified subgroup analysis, the effect of GTN (5 mg/day for 7 days) versus no-GTN was studied in 629 patients with intra-cerebral hemorrhage presenting within 48 hours and with systolic BP greater than or equal to 140 mm Hg. The primary outcome was the modified Rankin Scale at 90 days. Mean BP at baseline was 172/93 mm Hg and significantly lower (difference -7.5/-4.2 mm Hg; both $p \leq 0.05$) on day 1 in 310 patients allocated to GTN when compared with 319 randomized to no-GTN. No difference in the modified Rankin Scale was observed between those receiving GTN versus no-GTN (adjusted OR for worse outcome with GTN, 1.04; 95 % CI: 0.78 to 1.37; $p = 0.84$). In the subgroup of 61 patients randomized within 6 hours, GTN improved functional outcome with a shift in the modified Rankin Scale (OR, 0.22; 95 % CI: 0.07 to 0.69; $p = 0.001$). There was no significant difference in the rates of serious adverse events between GTN and no-GTN. The authors concluded that in patients with intra-cerebral hemorrhage within 48 hours of onset, GTN lowered BP and was safe; but did not improve functional outcome. They stated that very early treatment might be beneficial but needs assessment in further studies.

Cerebrolysin

Ziganshina and colleagues (2016) stated that cerebrolysin is a mixture of low-molecular-weight peptides and amino acids derived from porcine brain tissue (it includes (and not limited to) brain-derived neurotrophic factor (BDNF), glial cell line-derived neurotrophic factor (GDNF), nerve growth factor (NGF), and ciliary neurotrophic factor (CNTF)), which has potential neuroprotective and neurotrophic properties. It is widely used in the treatment of acute ischemic stroke in Russia, Eastern Europe, China, and other Asian and post-Soviet countries. In a Cochrane review, these researchers evaluated the benefits and risks of cerebrolysin for treating AIS. In May 2016 these investigators searched the Cochrane Stroke Group Trials Register, CENTRAL, Medline, Embase, Web of Science Core Collection, with Science Citation Index, LILACS, OpenGrey, and a number of Russian Databases. They also searched reference lists, ongoing trials registers and conference proceedings, and contacted the manufacturer of cerebrolysin, EVER Neuro Pharma GmbH (formerly Ebewe Pharma). Randomized controlled trials comparing cerebrolysin, started within 48 hours of stroke onset and continued for any time, with placebo or no treatment in people with AIS were selected for analysis. Two review authors independently applied inclusion criteria, assessed trial quality and risk of bias, and extracted data. They identified 6 RCTs (1,501 participants) that met the inclusion criteria. These investigators evaluated risk of bias and judged it to be unclear for generation of allocation sequence in 4 studies and low in 2 studies; unclear for allocation concealment in 5 studies and low in 1 study; high for incomplete outcome data (attrition bias) in 5 studies and unclear in 1 study; unclear for blinding; high for selective reporting in 4 studies and unclear in 2; and high for other sources of bias in 3 studies and unclear in the rest. The manufacturer of cerebrolysin, pharmaceutical company EVER Neuro Pharma, supported 3 multi-center studies, either totally, or providing cerebrolysin and placebo, randomization codes, research grants, or statisticians. None of the included trials reported on poor functional outcome defined as death or dependence at the end of the follow-up period or early death (within 2 weeks of stroke onset). All-cause death: these researchers extracted data from 5 trials (1,417 participants). There was no difference in the number of deaths: 46/714 (6.4 %) in cerebrolysin group versus 47/703 (6.7 %) in placebo group; RR 0.91 95 % CI 0.61 to 1.35 (5 trials, 1,417 participants, moderate-quality evidence). Serious adverse events (SAEs): 2 trials reported on this outcome, with 90 % confidence cerebrolysin increased the risks of SAEs by at least 1/3 compared to placebo: 62/589 (10.5 %) in cerebrolysin group versus 46/600

(7.7 %) in placebo group; RR 1.37 90 % CI: 1.01 to 1.86 (2 trials, 1,189 participants, moderate-quality evidence). Total number of people with AEs: 3 trials reported on this. There was no difference in the total number of people with AEs: 308/667 (46.2 %) in cerebrolisin group versus 307/668 (46.0 %) in placebo group; RR 0.97 95 % CI: 0.86 to 1.09, random-effects model (3 trials, 1,335 participants, moderate-quality evidence). The authors concluded that the findings of this Cochrane review did not demonstrate clinical benefits of cerebrolisin for treating AIS. These investigators found moderate-quality evidence suggesting that SAEs may be more common with cerebrolisin use in AIS.

In a meta-analysis, Wang and colleagues (2017) evaluated the clinical value of cerebrolisin and explored the potential influencing factors. The main electronic databases, including Medline, Embase, and the Cochrane Library, were searched. The primary outcome was functional recovery at Day 90. The secondary outcomes included mortality and adverse events (AEs). A total of 1,649 patients with AIS were pooled from 6 RCTs. Cerebrolisin had no significant effect on functional recovery at Day 90 compared with the effect of placebo as shown by the mRS response (RR 1.33, 95 % CI: 0.79 to 2.24, $p = 0.28$), NIHSS response (RR 1.03, 95 % CI: 0.83 to 1.28, $p = 0.77$), and Barthel Index (BI) response (RR 0.95, 95 % CI: 0.84 to 1.08, $p = 0.44$). In safety analysis, cerebrolisin did not increase the risk of AEs (RR 0.98, 95 % CI: 0.88 to 1.09, $p = 0.67$), risk of SAEs (RR 1.20, 95 % CI: 0.86 to 1.66, $p = 0.29$), or the mortality rate (RR 0.86, 95 % CI: 0.57 to 1.31, $p = 0.49$). The authors concluded that routine administration of cerebrolisin to patients with AIS cannot be supported by the available evidence from RCTs.

In a meta-analysis, Zhang and associates (2017) examined the safety and effectiveness of cerebrolisin for AIS. PubMed, Embase, and Cochrane Library were searched for RCTs, which intervened within 72 hours after the stroke onset. These researchers investigated the effectiveness and safety outcomes, respectively. Risk ratios and mean differences were pooled with fixed-effects model or random-effects model. A total of 7 studies were identified, involving 1,779 patients with AIS. The summary results failed to demonstrate significant superiority of cerebrolisin in the assessment of efficacy outcomes of mRS and BI. Similarly, use of cerebrolisin had neutral effects on safety outcomes compared with placebo, including mortality and SAEs. However, the number of included studies was small, especially in the analysis of efficacy outcomes, which might cause publication bias and inaccurate between-studies variance in the meta-analysis. The authors concluded that although cerebrolisin appeared to be safe, its routine use for the improvement of long-term rehabilitation following AIS is not supported by available evidence.

Erythropoietin

Yao and colleagues (2017) stated that erythropoietin (EPO) for treating AIS has been investigated in many studies. However, the evidence was inconsistent. These investigators performed a systematic review and meta-analysis to examine the role of EPO in treating patients with AIS. Two electronic databases (PubMed and Embase) were used; 30-day NIHSS measures primary outcome while all-cause mortality in the follow up and 90-day Barthel Index were regarded as secondary outcomes. Results were presented as RR, standardized mean difference (SMD) and 95 % CI. They employed Stata software to perform the meta-analysis. A total of 4 trials involving 784 patients were contained in this meta-analysis. The total combined results on 30-day NIHSS were (SMD = -0.52, 95 % CI: -1.39 to 0.34) with random-effects model and sensitivity analysis showed a significant difference after excluding the Ehrenreich 2009 trial. The total combined secondary measured results were (RR = 1.72, 95 % CI: 1.10 to 2.70) and (SMD = 0.01, 95 % CI: -0.14 to 0.16) for all-cause mortality and 90-day BI. In the subgroup analysis by whether used recombinant tPA (rtPA) earlier, the rtPA group showed increased all-cause mortality with the result of (RR = 1.92, 95 % CI: 1.04 to 3.52), but not in non-rtPA group. The authors concluded that the findings of this study showed no significant efficiency on the functional recovery after AIS with EPO and suggested an increased mortality in EPO group especially with combination of rtPA. They stated that large experimental studies particularly in non-human primates' models should be performed to investigate the safety and effectiveness of EPO and a combination with rtPA before conducting further clinical trials.

Normobaric Oxygen Treatment

Shi and colleagues (2016) stated that oxygen therapy has been shown to increase oxygen supply to ischemic tissues and improve outcomes after cerebral ischemia/re-perfusion. Normobaric hyperoxia (NBO), an easily applicable and non-invasive method, showed

protective effects on AIS animals and patients in pilot studies. However, many critical scientific questions are still unclear (e.g., the therapeutic time window of NBO, the long-term effects and the benefits of NBO in large clinic trials). These investigators reviewed the current literatures on NBO treatment of AIS in pre-clinical and clinical studies and analyzed and identified the key gaps or unknowns in the understanding of NBO.

The authors noted that NBO treatment in all published clinical trials was performed in hospital after the patients had reached there. The average starting time of NBO administration was 6 to 10 hours, or more, after onset of ischemia. These researchers stated that they look forward to the clinical trials designed for NBO treatment as early as in the ambulance; determining the therapeutic time window of NBO treatment is critical in clinic. It is important to determine how long after ischemia NBO treatment is still of beneficial effects. In addition, NBO pre-conditioning could be applied to TIA patients, who are at high risk of stroke and 1/3 of whom is likely to develop into AIS.

In addition, research on the efficacy of NBO in sub-acute and chronic patients is needed. The authors concluded that as a non-pharmaceutical and non-invasive treatment, NBO is worthy of notice as a promising AIS therapy; further pre-clinical and clinical studies are needed to determine its optimal therapy time window and long-term effects.

They stated that large clinical studies with optimal inclusion criteria of patients and the strategies of NBO treatment are needed.

Liu and Jin (2016) noted that the presence of a salvageable penumbra, a region of ischemic brain tissue with sufficient energy for short-term survival, has been widely agreed as the premise for thrombolytic therapy with tPA, which remains the only FDA-approved treatment for AIS. However, the use of tPA has been profoundly constrained due to its narrow therapeutic time window and the increased risk of potentially deadly hemorrhagic transformation (HT). Blood brain barrier (BBB) damage within the thrombolytic time window is an indicator for tPA-induced HT and both NBO and hypothermia have been shown to protect the BBB from ischemia/re-perfusion injury. Thus, providing oxygen as soon as possible (NBO treatment), freezing the brain (hypothermia treatment) to slow down ischemia-induced BBB damage or their combined use may extend the time window for the treatment of tPA. The authors summarized the protective effects of NBO, hypothermia or their use combined with tPA on ischemia stroke, based on which, the combination of NBO and hypothermia may be an ideal early stroke treatment to preserve the ischemic penumbra. These investigators stated that there is an urgent need to conduct large RCTs to address the effect.

Cheng et al (2021) noted that adjuvant neuro-protective therapies for AIS have demonstrated benefit in animal studies, albeit without human translation. In a prospective RCT, these investigators examined the safety and effectiveness of high-flow normo-baric oxygen (NBO) following endovascular re-canalization in anterior circulation stroke. Eligible patients were randomized to receive high-flow NBO by a Venturi mask (FiO₂ 50 %, flow 15 L/min) or routine low-flow oxygen supplementation by nasal cannula (flow 3 L/min) after vessel re-canalization for 6 hours. Patient demographics, procedural metrics, complications, functional outcomes, symptomatic ICH (sICH), and infarct volume were assessed. A total of 91 patients were treated with high-flow NBO. NBO treatment revealed a common OR of 2.2 (95 % CI: 1.26 to 3.87) favoring the distribution of global disability scores on the mRS at 90 days. The mortality at 90 days was significantly lower in the NBO group than in the control group, with an absolute difference of 13.86 % (RR of 0.35; 95 % CI: 0.13 to 0.93). A significant reduction of infarct volume as determined by MRI was noted in the NBO group. The median infarct volume was 9.4 ml versus 20.5 ml in the control group (beta coefficient of - 20.24; 95 % CI: - 35.93 to - 4.55). No significant differences were observed in the rate of sICH, pneumonia, urinary infection, and seizures between the 2 groups. The authors concluded that the findings of this study suggested that high-flow NBO therapy following endovascular re-canalization was safe and effective in improving functional outcomes, decreasing mortality, and reducing infarct volumes in anterior circulation stroke patients within 6 hours from stroke onset; vessel re-canalization before initiation of NBO may augment the effectiveness of NBO in providing neuroprotection following AIS.

The authors stated that this study had several drawbacks. First, this trial was performed in a single center. Second, high-flow NBO was administrated only after vessel re-canalization in this study and future studies may consider initiation of the high-flow NBO supplementation before re-perfusion therapy. Third, absence of adjusted infarct volumes based on baseline data may weaken the findings on neuroprotection of NBO to some degree. To comprehensively understand the beneficial effect of NBO, baseline diffusion-weighted imaging (DWI) with perfusion and collateral data, allowing for adjustment and more accurate assessment of infarct volumes will be considered in future studies (on this

note, the initial ASPECTS scores were similar between groups, as was the time from symptom onset to re-canalization; therefore, it was likely that the DWI data were accurate to some degree). Fourth, due to the concern of arterial blood gases may complicate this simple therapy, oxygen concentration in the blood was not monitored before or during the NBO process. However, it could not be ruled out that more patients possibly presenting with hypoxia were in the NBO arm. The data on the pulmonary post-endovascular therapy (EVT)/tPA course should have been analyzed in detail, in order to further determine the mechanisms underlying beneficial effect of NBO. The results of this study strongly suggested a therapeutic potential of NBO for AIS following endovascular re-canalization; thus, it warrants a multi-center study with large sample to demonstrate the generalization of this finding.

Mahmood et al (2022) stated that NBO is potentially a readily accessible neuro-protective therapy. In a systematic review and meta-analysis, these researchers examined the effectiveness of NBO in the treatment of acute stroke. Medline, Embase, and CENTRAL databases were searched to December 2020; RCTs of NBO administered less than 7 days after stroke to normoxic patients with no other indication for oxygen were identified. Data on early neurological recovery; functional outcome; mortality; oxygen saturation, and imaging markers were collected. A total of 15 publications involving 12 cohorts and 9,255 participants were identified. One study with 8,003 participants had low risk of bias, but the designs of smaller trials had limitations; 97 % of participants were in studies of low-flow oxygen (4 L/min or less); 82.8 % had ischemic stroke. Median time to treatment was 19.3 hours. Meta-analysis demonstrated no significant effect on: reduction in NIHSS at 7 days in all stroke or ischemic stroke only (MD of -0.16 [-1.11 to 0.80] and -0.73 [-3.54 to 2.08], respectively); mRS at 3 to 6 months of follow-up (combined SMD of -0.08 [-0.38 to 0.22]; 3 months SMD -0.01 [-0.03 to 0.029]; 6-month SMD -0.20 [-1.49 to 1.09]), or mortality (OR of 1.15 [0.87 to 1.53]). The majority of patients were administered low-flow oxygen in the sub-acute phase. Intervention strategies targeted at modification of early tissue survival (higher oxygen delivery and administration at early time-points when significant volumes of viable tissue persist) have not been tested adequately. The authors concluded that studies of NBO have shown no significant effect on early neurological recovery, functional outcome, or mortality in acute stroke. These investigators stated that oxygen has been predominantly low flow commenced in the sub-acute phase; further study of high-flow oxygen in the hyper-acute period may be warranted.

The authors stated that the main drawback of this systematic review and meta-analysis was the large proportion of subjects contributed by a single trial; thus, meta-analysis largely reflected the results of this trial. Analyses of other sub-groups such as hyper-acute populations or high-flow oxygen were limited by the small numbers in studies using such protocols. Detailed methodology was not available for several smaller clinical trials, some of which had a high risk of bias often due to lack of blinding in outcome assessment. These factors highlighted the lack of data for high-flow oxygen during the hyper-acute period after stroke onset.

Defibrinogen Therapy

Chen and colleagues (2018) examined the effectiveness of defibrinogen therapy on functional recovery and safety among 1,332 consecutive ischemic stroke patients who had not received IV thrombolysis with recombinant tPA (r-tPA). Stroke patients undergoing conservative and relatively individualized multiple-day dosing regimens of defibrinogen therapy between January 1, 2008 and May 30, 2016 were enrolled. Data were analyzed according to functional success (Barthel Index [BI] of 95 or 100, mRS of 0 or 1) and safety variables (intra-cranial hemorrhage [ICH], mortality and stroke recurrence). At 12 months, 18.62 % (203/1,087) of patients were lost to follow-up. The functional success rates were 39.84 % (526/1,320) and 42.23 % (459/1,087) as assessed by BI at 3 months and 12 months, respectively; 15 patients had asymptomatic ICH within 24 hours after the initial defibrase administration. During the 14 days after hospitalization, 12 patients were diagnosed with symptomatic ICH (sICH) and a total of 12 patients died from all causes. At 3 months, 56 patients were dead and 21 patients had recurrent stroke.

The percentage of death and recurrence of stroke at 12 months were 6.81 % and 3.22 %, respectively. Results from the historical control showed no significant differences of functional success were detected between the patients treated with r-tPA within 6 hours of stroke onset in NINDS II and the patients treated with defibrase within 6 hours after stroke in the present study. The authors concluded that the multiple-day dosing regimen of defibrinogen therapy using defibrase applied in the present study could achieve functional improvement among AIS patients, with low risks of mortality when compared with

other similar studies. However, they stated that RCTs with large sample size are needed to verify the safety and efficacy and safety of such a defibrinogenating therapy.

The authors stated that this study had several drawbacks. First, there were incomplete outcome data for 11.82 % of patients at 3 months and 18.68 % at 12 months because of loss to follow-up. Although analyses that compared the differences of demographic and baseline clinical characteristics between the patients failed to follow-up and patients followed-up at 3 months and 12 months identified no significant differences. Relatively high loss to follow-up may have influenced the evaluation of safety and effectiveness of multiple-day dosing regimen adopted in the current study. Second, because this study was limited due to the inherent weaknesses of a retrospective review, information such as stroke etiology and volume of stroke infarction lesion were not available. These potential confounders were not tried to be controlled by statistical analysis and may exert some influences on the results. Lastly, the present study was a case-series study, not a controlled trial, and thus there should be more caution in interpreting the results because of the lack of control group. Although the historical control was performed, the effects of such comparison were far more less than that of concurrent controls due to variable factors, such as the different participant cohort and different study conditions.

Microbubbles Combined with Ultrasound Sonothrombolysis

Auboire and colleagues (2018) stated that microbubbles (MBs) combined with ultrasound sonothrombolysis (STL) appeared to be an alternative therapeutic strategy for AIS, but clinical results remain controversial. In a systematic review, these investigators identified the parameters tested; evaluated evidence on the safety and efficacy on pre-clinical data on STL; and examined the validity and publication bias. PubMed and Web of Science databases were systematically searched from January 1995 to April 2017 in French and English. These investigators included studies evaluating STL on animal stroke model. This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Data were extracted following a pre-defined schedule by 2 of the authors. The CAMARADES criteria were used for quality assessment. A narrative synthesis was conducted. A total of 16 studies met the inclusion criteria. The result showed that ultrasound parameters and types of MBs were heterogeneous among studies. Numerous positive outcomes on efficacy were found, but only 4 studies demonstrated superiority of STL versus recombinant tPA on clinical criteria. Data available on safety were limited. The authors concluded that further in-vivo studies are needed to demonstrate better safety and efficacy of STL compared to currently approved therapeutic options. Moreover, these researchers stated that the future explorations on the safety of STL are essential. To achieve this objective, animal models, which reproduce the human pathophysiology (i.e., older animals, cardiovascular risk factors, significant duration of ischemia) should be used. Finally, MRI guidance and cavitation detection have not been evaluated in ischemic stroke yet but their use in STL look promising to guarantee a good safety/efficacy profile.

Guidelines from the American Heart Association on the early management of acute ischemic stroke (Powers, et al., 2019) state that the use of sonothrombolysis as adjuvant therapy with IV fibrinolysis is not recommended.

Microstent Retriever

Kuhn and associates (2017) evaluated the safety and efficacy of the “Baby Trevo” (Trevo XP ProVue 3x20 mm Retriever) stent retriever for large vessel occlusions (LVOs) in patients with AIS. These investigators retrospectively analyzed their stroke database and included all patients treated with the Baby Trevo for distal LVOs in AIS. Patient gender, mean age, vascular risk factors, NIHSS score at presentation, and mRS score at baseline and 90-day follow-up were documented. Re-perfusion rates for the vessels treated were recorded using the thrombolysis in cerebral infarction (TICI) classification. Occurrence of vasospasm and new or evolving infarcts in the treated vascular territory was documented. A total of 35 subjects with a mean NIHSS score of 18 were included. The Baby Trevo device was used in 38 branches of the anterior and posterior circulations; TICI 2b/3 blood flow was restored after 1 single pass in 20/38 (52.6 %) and after 2 or 3 passes in 11 vessels. The remaining vessels required either more than 3 passes, showed less than a TICI 2b/3 re-perfusion (n = 3), or demonstrated failure to retrieve the clot (n = 4); TICI 2b/3 re-perfusion was achieved in 30 patients (85.7 %). No vessel injuries, rupture, or significant vasospasm were observed. Overall, a mRS score of less than or equal to 2 was seen in 56.5 % of the subjects successfully

treated with the Baby Trevo at 90 days and in 81.3 % of surviving patients; 7 patients died (20 %). The authors concluded that these preliminary findings suggested that the “Baby Trevo” achieved a high recanalization rate without any significant risk. Moreover, they stated that larger cohort studies are needed to validate the clinical benefit.

Muller-Eschner and colleagues (2019) described their first experience using a small stent retriever specifically designed for thrombectomy in cerebral arteries with a small caliber (Acandis Aperio 3.5/28) in patients with AIS. All patients with an AIS, who underwent endovascular recanalization using the Aperio thrombectomy device with a diameter of 3.5 mm, were identified in retrospect and included in the present analysis.

Demographic and clinical data as well as data on the procedures performed were collected (patient sex, mean age, NIHSS, mRS, TIC1 score, and complications). Stent retriever-based thrombectomy with the Aperio 3.5/28 alone (n = 10 vessels) or in combination with other devices (n = 13 vessels) was performed in 22 AIS patients with embolic occlusions of distal branches of the anterior and posterior circulations (median NIHSS = 8.5). For vessels treated with the Aperio 3.5/28, these researchers achieved a TIC1 2b/3 reperfusion rate of 73.9 %; 1 patient suffered a symptomatic intra-cerebral hemorrhage after thrombectomy; otherwise, no procedure-related complications were observed. The authors concluded that these findings suggested that mechanical thrombectomy of distal cerebral artery occlusions with the Aperio 3.5/28 was feasible and in general safe, thus offering a promising option for endovascular stroke therapy. However, they stated that multi-center studies with larger patient cohorts are needed to evaluate the clinical benefit.

Minocycline

Malhotra and colleagues (2018) noted that various RCTs have investigated the neuroprotective role of minocycline in AIS or acute ICH patients. These researchers examined the safety and efficacy of minocycline in patients with acute stroke. These investigators carried out a literature search that spanned through November 30, 2017 across major databases to identify all RCTs that reported following efficacy outcomes among acute stroke patients treated with minocycline versus placebo: NIHSS, BI, and mRS scores. Additional safety, neuroimaging and biochemical end-points were extracted. They pooled MD and RR from RCTs using random-effects models. These researchers identified 7 RCTs comprising a total of 426 patients. Of these, additional unpublished data were obtained on contacting corresponding authors of 5 RCTs. In pooled analysis, minocycline demonstrated a favorable trend towards 3-month functional independence (mRS-scores of 0 to 2) (RR = 1.31; 95 % CI: 0.98 to 1.74, p = 0.06) and 3-month BI (mean difference [MD] = 6.92; 95 % CI: - 0.92 to 14.75; p = 0.08). In AIS subgroup, minocycline was associated with higher rates of 3-month mRS-scores of 0-2 (RR = 1.59; 95 % CI: 1.19 to 2.12, p = 0.002; I² = 58 %) and 3-month BI (MD = 12.37; 95 % CI: 5.60 to 19.14, p = 0.0003; I² = 47 %), whereas reduced the 3-month NIHSS (MD - 2.84; 95 % CI: - 5.55 to - 0.13; p = 0.04; I² = 86 %). Minocycline administration was not associated with an increased risk of mortality, recurrent stroke, MI and hemorrhagic conversion. The authors concluded that although data were limited, minocycline demonstrated efficacy and appeared a promising neuroprotective agent in acute stroke patients, especially in AIS subgroup. Moreover, they stated that further RCTs are needed to evaluate the safety and efficacy of minocycline among ICH patients.

Statins (e.g., Simvastatin)

In a randomized, double-blind, parallel, controlled trial, Uransilp and colleagues (2018) evaluated neurological outcomes and serum soluble LOX-1 (sLOX-1) and nitric oxide (NO) levels in patients with AIS treatment with low-dose 10 mg/day and high-dose 40 mg/day of simvastatin. A total of 65 patients with AIS within 24 hours after onset were randomized to treatment with simvastatin 10 mg/day or 40 mg/day for 90 days. Personal data and past history of all patients were recorded at baseline. The blood chemistries were measured by standard laboratory techniques. Serum sLOX-1 and NO levels and neurological outcomes including NIHSS, mRS, and BI were tested at baseline and day 90 after simvastatin therapy. Baseline characteristics were not significantly different in both groups except history of hypertension. Serum sLOX-1 and NO levels significantly reduce in both groups (sLOX-1 = 1.19 ± 0.47 and 0.98 ± 0.37 ng/ml; NO = 49.28 ± 7.21 and 46.59 ± 9.36 μ mol/L) in 10 mg/day and 40 mg/day simvastatin groups, respectively. Neurological outcomes including NIHSS, mRS, and BI significantly improve in both groups. However, no difference in NO level and neurological outcomes was found at 90 days after treatment as compared between low-dose 10 mg/day and high-dose 40 mg/day of simvastatin. The authors

concluded that high-dose simvastatin might be helpful to reduce serum sLOX-1. However, no difference in clinical outcomes was found between high- and low-dose simvastatin. These researchers stated that further more intensive clinical trials are needed to confirm the appropriate dosage of simvastatin in patients with AIS.

The authors stated that this study had several drawbacks. First, this study was cross-sectional, thereby allowing the determination of associations but not formulation of risk predictions. In addition, the study populations were relatively small ($n = 65$). Thus, these findings need further investigation in prospective studies with larger sample size.

Lastly, sLOX-1 and NO levels might be higher or lower in patients with intra-cranial atherosclerotic stenosis (ICAS) than in general population. Therefore, a normal control group should be included in future studies to evaluate the degree of impact of the presence and severity of AIS. The low proportion of patients with neurological progression could be secondary to a selection bias because of the admission of patients with less severe symptoms. Furthermore, neurological improvement in stroke patient could be from other factors than statin: age, NIHSS scale on admission, glycated hemoglobin (HbA1c) level, as well as location of stroke.

Golab-Janowska and associates (2018) stated that endothelial progenitor cells (EPCs) have been suggested to be a therapeutic option in AIS. Statins modulate endothelial function and preserve blood flow to tissue exposed to an ischemic insult. These researchers tested the hypothesis that statins therapy might augment circulating EPCs in patients with AIS. Demographic data, classical vascular risk factors, treatment and NIHSS data were prospectively collected from 43 consecutive AIS patients (group I), comprising 30 treated with statins (group Statin(+)) and 13 untreated (group Statin(-)). Risk factor controls (group II) included 22 subjects matched by age, gender, and traditional vascular risk factors. EPCs were measured by flow cytometry; and the populations of circulating stem cells (CD133+), early EPCs (CD133+/VEGFR2+) and ECs CD34⁺/CD133⁺/VEGFR2+ cells were analyzed. Patients ages ranged from 54 to 92 years (mean age of 75.2 ± 11.3 years). The number of CD34⁺/CD133⁺/VEGFR2+ cells was significantly lower in group I than II ($p < 0.05$). In group Statin(+) neurological deficit on the 1st, 3rd and 7th day was significantly lower in comparison Statin(-) ($p < 0.05$). These researchers observed significantly more frequent "improvement of greater than 50 % or complete recovery" and less frequent death in the statin-treated group. The number of early EPCs and ECs was significantly higher in the treated group on the 3rd day ($p < 0.05$). The authors concluded that statins treatment is likely to have a positive effect on spontaneous CD133+/VEGFR2+ and CD34⁺/CD133⁺/VEGFR2+ cell mobilization triggered by stroke. These preliminary findings need to be validated by well-designed studies.

Tenecteplase

Xu and co-workers (2018) stated that recent studies showed inconsistent results of tenecteplase versus alteplase for AIS with safety and efficacy. These researchers carried out a meta-analysis to examine the value of tenecteplase and alteplase in AIS treatment. Medline, Embase, and Cochrane Library from January 2001 to April 2018 were searched for RCTs with tenecteplase versus alteplase for AIS. The primary outcomes were early neurological improvement at 24 hours and functional outcome at 3 months. These investigators pooled 1,390 patients from 4 RCTs. Tenecteplase showed a significant early neurological improvement ($p = 0.035$) compared with alteplase. In addition, tenecteplase showed a neutral effect on excellent outcome ($p = 0.309$), good functional outcome ($p = 0.275$), and recanalization ($p = 0.3$). No significant differences in safety outcomes were demonstrated. In subgroup analysis, 0.25 mg/kg dose of tenecteplase showed a significantly increased early neurological improvement ($p < 0.001$). In serious stroke at baseline (NIHSS greater than 12) subgroup, tenecteplase showed a dramatic early neurological improvement ($p = 0.002$) and low risks of any intra-cranial hemorrhage (ICH) ($p = 0.027$). The authors concluded that tenecteplase provided better early neurological improvement than alteplase. The 0.25 mg/kg dose of tenecteplase subgroup specially showed better early neurological improvement and lower any ICH tendency than that of alteplase. In addition, in serious stroke at baseline subgroup, tenecteplase showed a lower risk of any ICH. They stated that these data provided guarantees for further researches on tenecteplase in AIS.

The authors stated that this study had several drawbacks. First, they performed this analysis based on limited data. Only 4 published RCTs with 1,390 patients were pooled to test the safety and efficacy of tenecteplase for AIS with different doses. Second, different RCT studies lacked uniform research indicators such as different doses of tenecteplase and disease severity in patients in these trials. In particular, the NOR-

TEST trial used high doses of tenecteplase and had a large sample size compared to other RCTs in relatively mild strokes. The data for all patients may be imbalanced and over-dominated by a single large trial with specific protocol conditions. Thus, the NOR-TEST trial caused a certain degree of bias to the conclusions. However, these investigators used different doses of subgroup analysis to reduce the bias. In addition, the NOR-TEST trial was a randomized, open-label, blinded end-point trial from 13 centers, which was better fit to the real situation. Third, only 1 RCT study was a phase-III clinical trial; hence, this meta-analysis lacked more comprehensive, multi-center, large-sample RCTs trials. Finally, in NOR-TEST9 clinical study, the AIS patients were milder and more people were lost to follow-up, which reduced the detection of outcome differences in intent-to-treat analysis.

Coutts and colleagues (2018) noted that alteplase has been the mainstay of thrombolytic treatment since the National Institutes of Neurological Disorders and Stroke trial was published in 1995. Over recent years, several trials have investigated alternative thrombolytic agents. Tenecteplase, a genetically engineered mutant tPA, has a longer half-life, allowing single IV bolus administration without infusion, is more fibrin specific, produces less systemic depletion of circulating fibrinogen, and is more resistant to plasminogen activator inhibitor compared to alteplase. Tenecteplase is established as the 1st-line IV thrombolytic drug for MI, where it has been shown to achieve comparable re-perfusion with reduced risk of systemic bleeding in comparison to alteplase. These investigators reviewed the literature on tenecteplase for the treatment of AIS, with a focus on the major completed and ongoing trials. The authors concluded that tenecteplase showed promise for treatment of AIS, both in populations currently eligible for alteplase and also in groups not currently treated with thrombolysis. They stated that trials are ongoing that are comparing tenecteplase with alteplase, and testing tenecteplase in subgroups of patients with ischemic stroke.

In a meta-analysis, Thelengana and associates (2019) examined if IV thrombolysis with tenecteplase in patients with AIS has better safety and efficacy outcomes than with IV alteplase. PubMed, Cochrane Central Register of Controlled Trials, WHO International clinical trials registry platform (ICTRP), Australian New Zealand Clinical Trials Registry (ANZCTR), EU Clinical Trials Register (EU-CTR) and ClinicalTrials.gov were searched for trials comparing tenecteplase with alteplase in AIS. Functional outcomes (mRS at 90 days), early major neurological improvement, rates of any ICH, symptomatic ICH and mortality rate at 90 days were the outcomes compared. A total of 4 RCTs involving 1,334 patients were included. The Tenecteplase group compared to the alteplase group had significantly better early major neurological improvement (RR = 1.56, 95 % CI: 1.00 to 2.43, $p = 0.05$). There was no significant difference between tenecteplase and alteplase in excellent functional outcome at 90 days, good functional outcome at 90 days, any ICH, ICH or mortality at 90 days. The authors concluded that this meta-analysis found tenecteplase to be significantly favoring one outcome: early major neurological improvement. Other outcomes did not differ between the tenecteplase and alteplase groups. Trials of cost-effective/benefit analysis comparing tenecteplase versus alteplase and tenecteplase versus endovascular treatment are needed to reinforce the evidence for the potential cost advantage of tenecteplase.

Ma et al (2022) stated that currently, studies regarding the safety and effectiveness of tenecteplase for the treatment of patients with AIS are still limited and inconsistent. In a systematic review and meta-analysis, these investigators compared the safety and effectiveness of tenecteplase with alteplase for the treatment of AIS patients. They carried out a literature search in PubMed, Embase, and Cochrane Library up to May 10, 2022. Primary outcomes of this study included 90-day good outcome (defined as an mRS score of 0 to 2) and 90-day excellent outcome (defined as an mRS score of 0 to 1); RRs with 95 % CIs were calculated using a random-effect model for each outcome. A total of 14 studies (3,537 patients) were included in this meta-analysis. There was no statistical difference between patients receiving tenecteplase and those receiving alteplase in the rates of 90-day good outcome (RR 1.01; 95 % CI: 0.91 to 1.13; $p = 0.79$) and 90-day excellent outcome (RR 1.04; 95 % CI: 0.92 to 1.19; $p = 0.50$). Patients receiving tenecteplase might associated with higher incidence of early neurologic improvement compared with those receiving alteplase (RR 1.29; 95 % CI: 1.04 to 1.61; $p = 0.02$). Furthermore, no statistical difference was observed between the 2 groups in other outcomes. The authors concluded that this meta-analysis indicated that tenecteplase in AIS patients was as safe and effective as alteplase and might provide more benefit than alteplase. However, due to several inherent limitations of this study, more prospective studies should be carried out to confirm these findings.

Katsanos et al (2022) noted that tenecteplase is being studied as an alternative thrombolytic agent for the treatment of AIS within ongoing RCTs. Furthermore, non-randomized clinical experiences with off-label use of tenecteplase versus alteplase for AIS treatment are being published. In a systematic review and meta-analysis, these researchers examined the available evidence on the safety and effectiveness of intravenous tenecteplase compared with intravenous alteplase provided by non-randomized studies. They identified eligible studies by searching Medline and Scopus databases. No language or other restrictions were imposed. The literature search was carried out on October 12, 2021. This meta-analysis used the PRISMA guidelines and was written according to the Meta-analysis of Observational Studies in Epidemiology (MOOSE) proposal. Non-randomized studies (prospective or retrospective) comparing intravenous tenecteplase (at any dose) with intravenous alteplase in patients with AIS were included in the analysis. The crude ORs and 95 % CIs were calculated for the association of tenecteplase versus alteplase with the outcomes of interest and adjusted ORs were extracted if provided. Estimates using random-effects models were pooled. The primary outcome was the probability of good functional outcome (mRS score of 0 to 2) at 90 days. A total of 6 studies were identified including a total of 1,820 patients (618 [34 %] treated with tenecteplase). Patients receiving tenecteplase had higher odds of 3-month good functional outcome (crude OR, 1.22; 95 % CI: 0.90 to 1.66; adjusted OR, 1.60, 95 % CI: 1.08 to 2.37), successful re-canalization (crude OR, 2.82; 95 % CI: 1.12 to 7.10; adjusted OR, 2.38; 95 % CI: 1.18 to 4.81), and early neurological improvement (crude OR, 4.88; 95 % CI: 2.03 to 11.71; adjusted OR, 7.60; 95 % CI: 1.97 to 29.41). No significant differences were detected in 3-month excellent functional outcome proportions (mRS score 0 to 1; crude OR, 1.53; 95 % CI: 0.81 to 2.91; adjusted OR, 2.51; 95 % CI: 0.66 to 9.49), symptomatic ICH (crude OR, 0.97; 95 % CI: 0.44 to 2.16; adjusted OR, 1.16; 95 % CI: 0.13 to 10.50), or parenchymal hematoma (crude OR, 1.20; 95 % CI: 0.24 to 5.95). The authors concluded that evidence from non-randomized studies suggested that tenecteplase is as safe as alteplase and potentially associated with improved functional outcomes compared with alteplase. Moreover, these researchers stated that based on these findings, enrollment in the ongoing RCTs appeared to be appropriate.

Investigational Therapies

An UpToDate review on “Approach to reperfusion therapy for acute ischemic stroke” (Filho and Samuels, 2019) states that “Investigational methods of reperfusion therapy for acute ischemic stroke include intra-arterial infusion of thrombolytic agents such as alteplase, ultrasound-enhanced thrombolysis, combined intravenous and intra-arterial thrombolysis, and glycoprotein IIb/IIIa antagonists such as tirofiban”.

Sphenopalatine Ganglion Stimulation

Bornstein and colleagues (2019) noted that sphenopalatine ganglion stimulation (SGS) increased cerebral collateral blood flow, stabilized the BBB, and reduced infarct size, in pre-clinical models of AIS, and showed potential benefit in a pilot randomized trial in humans. In a randomized, double-blind, sham-controlled study (the pivotal ImpACT-24B Trial), these researchers examined if SGS 8 to 24 hours after AIS would improve functional outcome. This study was carried out at 73 centers in 18 countries. It included patients (men aged 40 to 80 years and women aged 40 to 85 years) with anterior-circulation AIS, not undergoing re-perfusion therapy. Enrolled patients were randomly assigned via web-based randomization to receive active SGS (intervention group) or sham stimulation (sham-control group) 8 to 24 hours after stroke onset. Patients, clinical care providers, and all outcome assessors were masked to treatment allocation. The primary efficacy end-point was the difference between active and sham groups in the proportion of patients whose 3-month level of disability improved above expectations. This end-point was evaluated in the modified intention-to-treat (mITT) population (defined as all patients who received 1 active or sham treatment session) and the population with confirmed cortical involvement (CCI) and was analyzed using the Hochberg multi-step procedure (significance in both populations if $p < 0.05$ in both, and in one population if $p < 0.025$ in that one). Safety end-points at 3 months were all SAEs, SAEs related to implant placement or removal, SAEs related to stimulation, neurological deterioration, and mortality. All patients who underwent an attempted SGS or sham stimulator placement procedure were included in the safety analysis. Between June 10, 201, and March 7, 2018, a total of 1,078 patients were enrolled and randomly assigned to either the intervention or the sham-control group; 1,000 patients received at least 1 session of SGS or sham stimulation and entered the mITT population (481 [48 %] received SGS, 519 [52 %] were sham controls), among whom 520 (52 %) patients had CCI on imaging.

The proportion of patients in the mITT population whose 3-month disability level was better than expected was 49 % (234/481) in the intervention group versus 45 % (236/519) in the sham-control group (OR 1.14, 95 % CI: 0.89 to 1.46; $p = 0.31$). In the CCI population, the proportion was 50 % (121/244) in the intervention group versus 40 % (110/276) in the sham-control group (1.48, 1.05 to 2.10; $p = 0.0258$). There was an inverse U-shaped dose-response relationship between attained SGS intensity and the primary outcome in the CCI population: the proportion with favorable outcome increased from 40 % to 70 % at low-midrange intensity and decreased back to 40 % at high intensity stimulation ($p = 0.0034$). There were no differences in mortality or SAEs between the intervention group ($n = 536$) and the sham-control group ($n = 519$) in the safety population. The authors concluded that SGS was safe for patients with AIS 8 to 24 hours after onset, who were ineligible for thrombolytic therapy. These researchers stated that although not reaching significance, the trial's results supported that, among patients with imaging evidence of cortical involvement at presentation, SGS is likely to improve functional outcome.

Furthermore, an UpToDate review on "Initial assessment and management of acute stroke" (Filho and Mullen, 2020) does not mention sphenopalatine ganglion stimulation as a management option.

Tirofiban

In a systematic review and meta-analysis, Zhou and colleagues (2020) examined the safety and efficacy of tirofiban when used for AIS patients not undergoing ET. These researchers carried out an electronic search for English-language studies on PubMed, Scopus, Embase, and CENTRAL (Cochrane Central Register of Controlled Trials) data-bases up to July 31, 2019. All types of studies comparing tirofiban monotherapy or combined IV thrombolysis and tirofiban therapy with controls for AIS patients were included. A total of 6 studies were included in the review; 3 examined tirofiban monotherapy while 3 compared IV thrombolysis and tirofiban therapy with controls. Meta-analysis indicated that tirofiban monotherapy did not significantly increase the incidence of ICH (OR 1.14, 95 % CI: 0.72 to 1.82, $p = 0.57$; $I^2 = 0\%$), sICH (OR 0.52, 95 % CI: 0.09 to 3.03, $p = 0.46$; $I^2 = 0\%$) and mortality (OR 0.53, 95 % CI: 0.13 to 2.07, $p = 0.36$; $I^2 = 63\%$) in AIS patients. Similarly, this analysis indicated no significant increase in the rates of ICH (OR 0.82, 95 % CI: 0.33 to 2.07, $p = 0.68$; $I^2 = 0\%$), sICH (OR 0.91, 95 % CI: 0.16 to 5.16, $p = 0.91$; $I^2 = 0\%$) and mortality (OR 1.50, 95 % CI: 0.42 to 5.38, $p = 0.54$; $I^2 = 0\%$) in AIS patients treated with combined IV thrombolysis and tirofiban therapy. Meta-analysis for functional outcome was not possible. The authors concluded that tirofiban appeared to be safe when used following IV thrombolysis or as monotherapy in AIS patients; however, conclusions regarding improvement in functional improvement could not be drawn. These researchers stated that further trials with large sample size and homogenous methodology are needed to provide robust evidence.

Gong and co-workers (2020) stated that the safety and efficacy of tirofiban for patients with AIS remains controversial. Thus, these investigators carried out a systematic review and meta-analysis. They searched PubMed, EmbaseE, Cochrane Library, Web of Science, and related international clinical trials registries through March 31, 2019, using the terms "tirofiban" and "stroke". All apparently unconfounded RCTs and cohort studies with 2 arms comparing treatment with and without tirofiban for AIS were included in this review. Primary outcomes included sICH, fatal ICH, mortality, and mRS (0 to 2) at 3 months. A total of 17 studies including 2,914 AIS patients were identified. Pooled results showed that tirofiban treatment in AIS did not increase the risk of sICH (OR, 0.95; 95 % CI: 0.71 to 1.28; $p = 0.75$) or mortality (OR, 0.80; 95 % CI: 0.64 to 1.02; $p = 0.07$). However, fatal ICH increased significantly in the tirofiban treatment group (OR, 2.84; 95 % CI: 1.38 to 5.85; $p = 0.005$), and subgroup analysis showed that tirofiban via IA administration was associated with increased risk of fatal ICH (OR, 2.90; 95 % CI: 1.12 to 7.55; $p = 0.03$), while IV administration was not (OR, 2.75; 95 % CI: 0.92 to 8.20; $p = 0.07$). Furthermore, tirofiban showed no obvious improvement in functional outcome (mRS 0 to 2) (OR, 1.29; 95 % CI: 0.97 to 1.71; $p = 0.08$). The authors concluded that tirofiban appeared to be safe in systemic treatment and may represent a potential choice for management of AIS. However, IA administration requires further adequately controlled studies in order to develop an appropriate protocol, similar to that in cardiology.

In a meta-analysis, Fu and associates (2020) examined the safety and efficacy of tirofiban compared with those without tirofiban in AIS patients receiving ET. These investigators carried out a systematic literature search in PubMed and Embase data-bases

without language or time limitation. Safety outcomes were sICH and mortality. Efficacy outcomes were re-canalization rate and favorable functional outcome. Review Manager 5.3 and Stata Software Package 15.0 were used to perform the meta-analysis. A total of 11 studies with 2,028 patients were included; 704 (34.7 %) patients were administered tirofiban combined with ET. Meta-analysis suggested that tirofiban did not increase the risk of sICH (OR 1.08; 95 % CI: 0.81 to 1.46; $p = 0.59$) but significantly decreased mortality (OR 0.68; 95 % CI: 0.52 to 0.89; $p = 0.005$). There was no association between tirofiban and re-canalization rate (OR 1.26; 95 % CI: 0.86 to 1.82; $p = 0.23$) or favorable functional outcome (OR 1.21; 95 % CI: 0.88 to 1.68; $p = 0.24$). Subgroup analyses indicated that pre-operative tirofiban significantly increased re-canalization rate (OR 3.89; 95 % CI: 1.70 to 8.93; $p = 0.001$) and improve favorable functional outcome (OR 2.30; 95 % CI: 1.15 to 4.60; $p = 0.02$). The authors concluded that tirofiban was safe in AIS patients with ET and could significantly reduce mortality; pre-operative tirofiban may be effective, but further studies are needed to confirm the efficacy.

Acute Stenting and Concomitant Tirofiban Administration

Bernava and colleagues (2021) noted that intracranial artery dissection (IAD) is an uncommon cause of AIS. Although acute stenting of the dissected arterial segment is a therapeutic option, the associated antiplatelet regimen remains a matter of debate. In a retrospective, single-center study, these researchers examined the safety and efficacy of acute intracranial stenting together with concomitant intravenous (IV) administration of tirofiban. These investigators also carried out a systematic review of the literature. These researchers reviewed the clinical and radiological records of all patients treated at their center by intracranial stenting in the setting of AIS between January 2010 and December 2020. They conducted a systematic review of the literature according to the PRISMA-P guidelines for relevant publications from January 1976 to December 2020 on IAD treated by stenting. A total of 7 patients with IAD underwent acute stenting with concomitant tirofiban during the study period. Mid-term follow-up showed parent artery patency in 6/7 cases (85.7 %). The mRS was less than or equal to 0 to 2 at 3 months in 5/7 cases (71.4 %). The literature review identified 22 patients with IAD treated with acute stenting in association with different anti-thrombotic therapies. Complete re-vascularization was attained in 86.3 % of cases with a mRS of less than or equal to 0 to 2 in 68 % of patients at 3-month follow-up. The authors concluded that the findings of this study suggested that intracranial stenting of IAD with concomitant tirofiban administration in the acute phase was a safe and effective endovascular treatment in patients with a small ischemic core.

The authors stated that this study had several drawbacks related to its retrospective design, single-center setting and the small number of patients ($n = 7$). Furthermore, tirofiban is not available in all countries; therefore, limiting the generalization of the findings. However, it should be noted that the rarity of IAD published reports has so far precluded a definition of a standard of care for such a disease and further research is needed.

Guidelines from the American Heart Association (Powers, et al., 2019) state that The safety and efficacy of IV glycoprotein IIb/IIIa inhibitors administered during endovascular stroke treatment are uncertain.

Governor Vessel Acupuncture

Shao and colleagues (2021) noted that AIS is the major type of stroke, which highly risks human health and quality of life (QOL). Governor vessel acupuncture (GV Ac) is a specific acupoint selection treatment. In a systematic review and meta-analysis, these researchers examined the clinical value of GV Ac in AIS patients. A total of 7 electronic databases were searched for all related RCTs before December 2020. The included studies met the following criteria: all target patients were diagnosed as AIS; the experimental group used GV Ac as the only intervention or combined with routine neurological therapy as conventional treatment; the control group received ordinary acupuncture, or the same conventional treatment as the experimental group, or both. A total of 18 studies were included, involving 1,543 AIS patients. The results showed that compared to the conventional treatment, GV Ac combining with conventional therapy resulted in BI (MD = 14.16, 95 % CI: 7.34 to 20.79) improvement, mRS (MD = -0.63, 95 % CI: -0.95 to -0.32, $p < 0.0001$) decrease, better NIHSS scores (MD = -1.18, 95 % CI: -1.52 to -0.83), and lower China Stroke Scale (CSS)/Modified Edinburgh-Scandinavia Stroke Scale (MESSS) scores (MD = -3.77, 95 % CI: -4.98 to -2.57). Furthermore, GV Ac could better improve activities of daily living (ADL) (MD = 8.27, 95 % CI: 4.29 to 12.26) and

neurological deficit scores (NIHSS: MD = -1.32, 95 % CI: -2.18 to -0.47; CSS/MESSS: MD = -4.63, 95 % CI: -5.91 to -3.35), when compared to the ordinary acupuncture. The authors concluded that according to the current evidence, GV Ac for AIS's efficacy appeared to be better than that of ordinary acupuncture. When combined with conventional treatment, GV Ac may increase the benefit. Moreover, these researchers stated that limited by the methodological quality of the included studies; they stated that larger, more high-quality, strictly designed RCTs in China and abroad are needed to confirm these findings.

The authors stated that this study had 2 main drawbacks. First, the included clinical trials were all carried out in China; thus, the findings may not be generalized to other races. Second, in the aspect of blind design, only 1 trial reported single-blind (blind to evaluators). The double-blind trial of acupuncture research is still controversial. Sham-acupuncture may have potential physiological effects. Furthermore, it may be difficult to perform acupuncture studies completely blind in some countries, such as China, because patients may be very familiar with the feeling of "de qi". Furthermore, not all subjects are willing to receive sham acupuncture.

Mechanical Thrombectomy With Intravenous Thrombolysis

Guidelines from the American Heart Association on the early management of persons with acute ischemic stroke (Powers, et al., 2019) state that patients eligible for IV alteplase should receive IV alteplase even if mechanical thrombectomy is being considered. In patients under consideration for mechanical thrombectomy, observation after IV alteplase to assess for clinical response should not be performed.

Yang and co-workers (2020) noted that in AIS, there is uncertainty regarding the benefit and risk of administering IV alteplase before endovascular thrombectomy. These researchers carried out a trial at 41 academic tertiary care centers in China to examine endovascular thrombectomy with or without IV alteplase in patients with AIS. Patients with AIS from large-vessel occlusion in the anterior circulation were randomly assigned in a 1:1 ratio to undergo endovascular thrombectomy alone (thrombectomy-alone group) or endovascular thrombectomy preceded by IV alteplase, at a dose of 0.9 mg/kg of body weight, administered within 4.5 hours after symptom onset (combination-therapy group). The primary analysis for non-inferiority assessed the between-group difference in the distribution of the mRS scores (range of 0 [no symptoms] to 6 [death]) at 90 days on the basis of a lower boundary of the 95 % CI of the adjusted common odds ratio (OR) equal to or larger than 0.8. These investigators evaluated various secondary outcomes, including death and re-perfusion of the ischemic area. Of 1,586 patients screened, 656 were enrolled, with 327 patients assigned to the thrombectomy-alone group and 329 assigned to the combination-therapy group. Endovascular thrombectomy alone was non-inferior to combined IV alteplase and endovascular thrombectomy with regard to the primary outcome (adjusted common OR, 1.07; 95 % CI: 0.81 to 1.40; $p = 0.04$ for non-inferiority) but was associated with lower percentages of patients with successful re-perfusion before thrombectomy (2.4 % versus 7.0 %) and overall successful re-perfusion (79.4 % versus 84.5 %). Mortality at 90 days was 17.7 % in the thrombectomy-alone group and 18.8 % in the combination-therapy group. The authors concluded that in Chinese patients with AIS from large-vessel occlusion, endovascular thrombectomy alone was non-inferior with regard to functional outcome, within a 20 % margin of confidence, to endovascular thrombectomy preceded by IV alteplase administered within 4.5 hours after symptom onset.

Suzuki and colleagues (2021) stated that whether IV thrombolysis is needed in combination with mechanical thrombectomy in patients with acute large vessel occlusion stroke is unclear. In a randomized, open-label, multi-center clinical trial, these researchers examined if mechanical thrombectomy alone is non-inferior to combined IV thrombolysis plus mechanical thrombectomy for favorable post-stroke outcome. This trial included 204 patients with AIS due to large vessel occlusion enrolled at 23 hospital networks in Japan from January 1, 2017 to July 31, 2019, with final follow-up on October 31, 2019. Patients were randomly assigned to mechanical thrombectomy alone ($n = 101$) or combined IV thrombolysis (alteplase at a 0.6-mg/kg dose) plus mechanical thrombectomy ($n = 103$). The primary efficacy endpoint was a favorable outcome defined as a mRS score (range of 0 [no symptoms] to 6 [death]) of 0 to 2 at 90 days, with a non-inferiority margin OR of 0.74, evaluated using a 1-sided significance threshold of 0.025 (97.5 % CI). There were 7 pre-specified secondary efficacy endpoints, including mortality by day 90. There were 4 pre-specified safety endpoints, including any ICH and symptomatic ICH within 36 hours. Among 204 patients (median age of 74 years; 62.7 % men; median NIHSS score of 18), all patients completed the trial. Favorable outcome occurred in 60 patients (59.4 %) in the mechanical thrombectomy alone group and 59 patients (57.3 %) in the combined IV

thrombolysis plus mechanical thrombectomy group, with no significant between-group difference (difference, 2.1 % [1-sided 97.5 % CI: -11.4 % to ∞]; OR, 1.09 [1-sided 97.5 % CI: 0.63 to ∞]; $p = 0.18$ for non-inferiority). Among the 7 secondary efficacy endpoints and 4 safety endpoints, 10 were not significantly different, including mortality at 90 days (8 [7.9 %] versus 9 [8.7 %]; difference, -0.8 % [95 % CI: -9.5 % to 7.8 %]; OR, 0.90 [95 % CI: 0.33 to 2.43]; $p > 0.99$). Any ICH was observed less frequently in the mechanical thrombectomy alone group than in the combined group (34 [33.7 %] versus 52 [50.5 %]; difference, -16.8 % [95 % CI: -32.1 % to -1.6 %]; OR, 0.50 [95 % CI: 0.28 to 0.88]; $p = 0.02$). Symptomatic ICH was not significantly different between groups (6 [5.9 %] versus 8 [7.7 %]; difference, -1.8 % [95 % CI: -9.7 % to 6.1 %]; OR, 0.75 [95 % CI: 0.25 to 2.24]; $p = 0.78$). The authors concluded that among patients with acute large vessel occlusion stroke, mechanical thrombectomy alone, compared with combined IV thrombolysis plus mechanical thrombectomy, failed to demonstrate non-inferiority regarding favorable functional outcome. However, the wide CIs around the effect estimate also did not allow a conclusion of inferiority.

The authors stated that this study had several drawbacks. First, this was an open-label study regarding the use of rt-PA. Second, the sample size and the non-inferiority margin of 0.74 were calculated from previous studies using a dose of 0.9 mg/kg of alteplase. Furthermore, the non-inferiority margin was selected using the fixed-margin method rather than the minimal clinically important difference. Third, favorable outcomes were more frequent than expected, which may have resulted in a reduction of study power. Recently, the Direct-MT trial in China, which was a similar randomized clinical trial to this trial, demonstrated non-inferiority for mechanical thrombectomy alone compared with combined IV thrombolysis and mechanical thrombectomy with regard to functional outcome. In addition, 3 randomized clinical trials (MR CLEAN-NO IV [ISRCTN80619088], SWIFT DIRECT [NCT03192332], and DIRECT-SAFE [NCT03494920]) similar to this trial are ongoing. Meta-analyses that include these trials may provide greater clarity.

Zi and associates (2021) noted that for patients with large vessel occlusion strokes, it is unclear if endovascular treatment alone compared with IV thrombolysis plus endovascular treatment (standard treatment) could achieve similar functional outcomes. In a randomized, multi-center, non-inferiority trial, these investigators examined if endovascular thrombectomy alone is non-inferior to IV alteplase followed by endovascular thrombectomy for achieving functional independence at 90 days among patients with large vessel occlusion stroke. This trial was carried out in 33 stroke centers in China. Patients ($n = 234$) were 18 years or older with proximal anterior circulation intracranial occlusion strokes within 4.5 hours from symptoms onset and eligible for IV thrombolysis. Enrollment took place from May 20, 2018 to May 2, 2020. Patients were enrolled and followed-up for 90 days (final follow-up was July 22, 2020). A total of 116 patients were randomized to the endovascular thrombectomy alone group and 118 patients to combined IV thrombolysis and endovascular thrombectomy group. The primary endpoint was the proportion of patients achieving functional independence at 90 days (defined as score 0 to 2 on the mRS; range of 0 [no symptoms] to 6 [death]). The non-inferiority margin was -10 %. Safety outcomes included the incidence of symptomatic ICH within 48 hours and 90-day mortality. The trial was stopped early because of efficacy when 234 of a planned 970 patients had undergone randomization. All 234 patients who were randomized (mean age of 68 years; 102 women [43.6 %]) completed the trial. At the 90-day follow-up, 63 patients (54.3 %) in the endovascular thrombectomy alone group versus 55 (46.6 %) in the combined treatment group achieved functional independence at the 90-day follow-up (difference, 7.7 %, 1-sided 97.5 % CI: -5.1 % to ∞) (p for non-inferiority = 0.003). No significant between-group differences were detected in symptomatic ICH (6.1 % versus 6.8 %; difference, -0.8 %; 95 % CI: -7.1 % to 5.6 %) and 90-day mortality (17.2 % versus 17.8 %; difference, -0.5 %; 95 % CI: -10.3 % to 9.2 %). The authors concluded that among patients with ischemic stroke due to proximal anterior circulation occlusion within 4.5 hours from onset, endovascular treatment alone, compared with IV alteplase plus endovascular treatment, met the pre-specified statistical threshold for non-inferiority for the outcome of 90-day functional independence. These findings should be interpreted in the context of the clinical acceptability of the selected non-inferiority threshold.

The authors stated that this study had several drawbacks. First, the non-inferiority margin used in this trial was not selected using the minimal clinically important difference or fixed margin methods. The non-inferior margin of 10 % was broad, reaching no consensus in the clinical community, which may lead to concerns regarding the robustness of study results. Second, the early termination of the trial created potential for over-estimation of the effect size; however, it followed a highly conservative pre-specified statistical stopping rule. Third, the median time from

hospital arrival to the start of IV alteplase of 61 mins (inter-quartile range [IQR], 49 to 81 mins) in this trial was similar to the 59 minutes (IQR, 45 to 78 mins) in the DIRECT-MT trial; but was longer than what has been reported in the other recent thrombectomy trials. The main reason for these differences was that alteplase could not be administered immediately after standard clinical and neuroimaging screening as they were in the previous trials because additional time was needed for study screening, obtaining patient consent, and randomization in this and the DIRECT-MT trials. This explanation was supported by the findings of a recent multi-national neuroprotection randomized trial. Despite having one of most efficient workflows for IVT and thrombectomy reported to-date, the median time from hospital arrival to the initiation of the experimental drug in the Safety and Efficacy of Nerinetide in Subjects Undergoing Endovascular Thrombectomy for Stroke (ESCAPE NA-1) trial was 64 mins (IQR, 48 to 86 mins). This highlighted that achieving hospital arrival to initiate IV alteplase times as short as those in the previous trials with the absence of waiver of informed consent was unlikely. In China, informed consent is required even before the standard administration of IVT because patients or their families often have to pay in advance before receiving alteplase. Relatedly, the median door-to-needle time was 62 mins (IQR, 40 to 94 mins) among 24,542 patients treated with IV alteplase in a recent analysis of the Chinese Stroke Center Alliance Program, a finding that was fully consistent with the times observed in this and the DIRECT-MT trials. Nevertheless, given the relatively long times from hospital arrival to alteplase initiation in this trial, it remained possible that IV alteplase might still be beneficial in the presence of faster delivery times. Moreover, as bridging with tenecteplase versus alteplase was associated with higher rates of pre-thrombectomy re-perfusion and better 90-day functional outcomes in the Tenecteplase Versus Alteplase Before Endovascular Therapy for Ischemic Stroke (EXTEND-IA-TNK) trial, it was possible that tenecteplase bridging might yield better results than thrombectomy alone. Fourth, because this, the DIRECT-MT, and the SKIP trials only included Asian patients, the generalizability of their findings was somewhat limited due to the significantly higher incidence of intra-cranial atherosclerotic disease in Asian than in Western populations.

Fingolimod Plus Standardized Treatment

Bai et al (2022) stated that AIS is the most common type of stroke. Fingolimod is a sphingosine analog that acts on sphingosine-1-phosphate receptors (S1PR). Recently, the safety and effectiveness of fingolimod in both patients with ICH and patients with AIS have been examined in proof-of-concept trials. In a systematic review and meta-analysis, these investigators examined the safety and effectiveness of fingolimod for AIS. This study was conducted according to the PRISMA guidelines. They searched for publications on the PubMed, Embase, Cochrane Central Register of Controlled Trials, Clinical trials, CNKI, Wanfang Data, VIP, CBM up to August 2021. These researchers compiled 5 studies; a main meta-analysis forest plots were conducted for the values of the proportion of patients whose mRS score was 0 to 1 at day 90. There were heterogeneities in each study; the method of sensitivity analysis was performed. A sensitivity analysis was performed with a MD of the effectiveness of fingolimod plus standardized treatment versus standardized treatment alone. Random effect model is used for meta-analysis regardless of the I2 index. The analysis was performed for categorical variables using the RR, LogRR, and its 95 % CI. The methodological quality of each RCTs was evaluated according to the Cochrane Collaboration tool to assess the risk of bias (ROB). A meta-analysis of 5 studies with 228 subjects was carried out. The RR of patients whose mRS score was 0 to 1 at day 90 between fingolimod plus standardized treatment and standardized treatment alone was 2.59 (95 % CI: 1.48 to 4.56). The Fingolimod plus standard treatment group decreased infarct growth and improved clinical function than the standard treatment. These researchers stated that this study showed that these early findings are promising; moreover, larger studies in different patient populations are needed to validate these findings.

The authors stated that this study had many drawbacks. First, although the search strategy was relatively complete, it did not rule out that eligible articles were not included. Second, a large sample of studies lacked in the included studies. Third, it only includes RCTs. Fourth, It did not distinguished patients who receive different standard treatments such as the intravenous administration of tPA, intravascular therapy, antiplatelet drugs, and so on. Fifth, 4 of the included trials came from the same group of investigators. Sixth, 4 of the excluded studies' data were not extractable. The records with unobtainable data may cause bias in the results. Seventh, none of the included trials was double-blinded (most had a PROBE design). (eight, high heterogeneity

across studies should not be neglected, although a random-effects model was used for adjustment. Nonetheless, results were broadly similar even if sensitivity analysis which decreased the heterogeneity were performed. Inherent limitations in the majority of meta-analyses, such as lack of access to raw data and the variety in definitions of outcomes in the included studies, were unavoidable. None of the included studies was adequately sized to evaluate the proposed primary endpoint. Ninth, the entire data were derived from patients in China. More studies are needed that include other ethnic groups. Tenth, different inclusion/exclusion criteria and follow-up periods in the included studies led to high heterogeneity. Eleventh, the treatment of 5 included studies did not cover intravascular therapy. Twelve, 5 studies included only phase-I or early phase-II clinical trials with very small sample sizes. Thirteen, the quality of studies was poor. Fourteen, the sample sizes for each of the analysis were small. Fifteen, 4 of 5 studies were from the same group and all from China. RCTs with greater patient numbers are needed for future studies. Sixteen, the method for infarct growth measured in the trials was not uniform.

Mesenchymal Stem Cells for Enhancement of Ischemic Stroke Recovery

Behzadifard et al (2023) stated that it has been reported that the therapeutic potential of stem cells is mainly mediated by their paracrine factors. In a systematic review, these investigators identified the effects of conditioned medium of mesenchymal stem cells (MSC-CM) against stroke. They searched PubMed, Scopus, and ISI Web of Science databases for all available studies relevant to the effects of MSC-CM against the middle cerebral artery occlusion (MCAO) model of ischemic stroke until August 2022. The quality of the included studies was examined using The STAIR scale. During the systematic search, a total of 356 published studies were identified. A total of 15 datasets were included following screening for eligibility. The type of cerebral ischemia was the MCAO model and CM was obtained from MSCs. The results showed that the therapeutic time window can be considered a crucial factor when researchers use MSC-CM for stroke therapy. Furthermore, MSC-CM therapy contributed to functional recovery and reduced infarct volume following stroke by targeting different cellular signaling pathways. The authors concluded that these findings demonstrated that MSC-CM therapy has the ability to improve functional recovery and attenuate brain infarct volume following ischemic stroke in pre-clinical studies. These researchers hope their study would accelerate needed progress towards clinical trials.

Percutaneous Transluminal Angioplasty with Stenting for the Treatment of Basilar Artery Stenosis

Machado et al (2022) noted that percutaneous transluminal angioplasty and stenting (PTAS) in acute stroke due to severe basilar artery stenosis (BAS) or basilar artery occlusion (BAO) remain a matter of debate. The higher risk of stroke recurrence in patients with vertebra-basilar stenosis compared to anterior circulation atherosclerotic disease creates high expectations concerning endovascular approaches. In a single-center study, these researchers reviewed their experience with PTAS in acute stroke caused by BAS/BAO. These investigators screened their prospective database from June 2014 until December 2020; and patients with acutely symptomatic severe (greater than 80 %) BAS or acute BAO who underwent PTAS were analyzed. A total of 25 patients included: 72 % men (mean age of 68.6 years), all with prior mRS of less than 2; 12 presented with acute BAO and were submitted to mechanical thrombectomy before PTAS, while the remaining had severe BAS. Successful stent placement was achieved in 22 (88 %). Procedure-related complications included new small ischemic lesions (16 %), basilar artery dissection (8 %), vertebral artery dissection (12 %) and death (12 %). At 3 months post-PTAS, 10 out of 23 patients (43.5 %) were independent (mRS of 2 or less) and 6 died; 14 patients underwent transcranial Doppler (TCD) ultrasound 3 months post-PTAS: 12 showed residual stenosis, 1 significant stent re-stenosis and 1 presented stent occlusion. The authors concluded that PTAS showed to be a technically feasible and reasonably safe procedure in selected patients; however, good clinical outcomes may be difficult to achieve as only 43.5 % of the patients remained independent at 3 months. These researchers stated that randomized studies are needed to confirm the safety and effectiveness of PTAS in acute stroke caused by basilar artery steno-occlusive disease.

Palmisciano et al (2023) stated that BAS carries high morbidity and mortality, with variable outcomes after endovascular treatments. In a systematic review, these investigators examined the literature on PTAS for the treatment of BAS. PubMed, Embase, Web-of-Science, Scopus, and Cochrane were searched using the PRISMA guidelines to include

prospective/retrospective cohort studies describing PTAS for BAS. Pooled rates of intervention-related complications and outcomes were analyzed with random-effect model meta-analyses. They included 25 retrospective cohort studies comprising 1,016 patients. All patients were symptomatic, presenting with transient ischemic attack (TIA) or ischemic stroke. BAS frequently involved the middle basilar artery (51.4 %), mostly classified as Mori-B (57.4 %). PTAS for BAS was indicated in severe (greater than or equal to 50 % to 70 %), symptomatic BAS refractory to dual anti-platelet therapy. Patients underwent angioplasty (95.5 %) and/or stenting (92.2 %), preferably using Wingspan or Apollo stents. Median baseline BAS was 81 % (range of 53 % to 99 %), while median post-intervention BAS was 13 % (0 % to 75 %). Actuarial rates of successful intervention and "good" final outcome were 100 % (95 % CI: 100 % to 100 %) and 89 % (95 % CI: 85 % to 93 %). Intervention-related recurrent ischemic stroke occurred in 85 patients (8.3 %) with actuarial rates of 5 % (95 % CI: 4 % to 7 %), differentiated into perforator (5.4 %), in-stent (2.6 %), and embolic (0.4 %). Actuarial rates of intervention-related dissection, re-stenosis, and death were 0 % (95 % CI: 0 % to 0 %), 1 % (95 % CI: 0 % to 1 %), and 0 % (95 % CI: 0 % to 2 %). The authors concluded that elective PTAS appeared to be safe and effective in selected patients with medically refractory, severe, symptomatic, and non-acute BAS. Different stent types and angioplasty-assisted procedures should be considered based on specific clinico-radiological characteristics of the lesions. Moreover, these researchers stated that future RCTs are needed to corroborate these findings.

In a retrospective, single-center study, Zhou et al (2023) examined the safety, effectiveness, and risk factors of endovascular re-canalization of symptomatic non-acute occlusion of the vertebra-basilar artery (SNOVA). Patients with SNOVA were enrolled and treated with endovascular re-canalization. The clinical data, endovascular treatment, peri-procedural complications, and follow-up outcomes were analyzed. A total of 88 patients were enrolled, with an interval to re-canalization of 2 to 89 days (median of 23 days) and an mRS of 2 to 5 (median of 3 and inter-quartile range [IQR] 1). Occlusion was in the intra-cranial vertebral artery in 68 (77.27 %) patients and basilar artery in 20 (22.73 %), with an occlusion length of 4.5 to 43.7 mm (mean of 18.3 ± 8.8 mm). Endovascular re-canalization was successful in 81 (92.0 %) patients. Post-dilatation was carried out in 23 (28.4 %) patients. After stenting, the residual stenosis was 10 % to 40 % (mean of 20.2 ± 7.6 %). Peri-procedural complications occurred in 17 (19.3 %) patients, with a mortality rate of 5.7 %. In total, 79 (95.18 %) patients underwent follow-up 5 to 29 (mean of 16.9 ± 5.5) months later, with an mRS score of 0 to 6 (median 1 and IQR 1) at follow-up, being significantly ($p < 0.0001$) better than that at discharge. Stroke occurred in 9 patients (11.4 %) in 1 year. In-stent re-stenosis occurred in 19 (25.33 %) patients. Significant ($p < 0.05$) independent risk factors were blunt occlusion for successful re-canalization, duration to re-canalization and blunt occlusion for peri-procedural complications, and post-dilatation for both in-stent restenosis and 1-year stroke or death events. The authors concluded that endovascular re-canalization of symptomatic non-acute occlusion of the vertebra-basilar artery was feasible even for a long occlusion segment, with a high re-canalization rate, a low complication rate, and a good prognosis. Blunt occlusion and duration from the onset to re-canalization may affect successful re-canalization and peri-procedural complications while post-dilatation may affect in-stent re-stenosis and prognosis.

The authors stated that this study had several drawbacks, including the retrospective and single-center study design, only Chinese patients enrolled, a small cohort of patients ($n = 88$), no comparison with symptomatic non-acute occlusion in the anterior circulation, and no randomization, which may all affect the generalization of the outcomes. These researchers stated that in the future, large, multi-center RCTs with different racial groups are needed to confirm these findings.

An UpToDate chapter on approaches to reperfusion therapy in acute ischemic stroke (Oliveira-Filho and Samuels, 2023) stated that "all eligible patients with acute ischemic stroke should be treated with intravenous thrombolysis, including those with stroke in the posterior circulation. Mechanical thrombectomy is beneficial for select patients with acute ischemic stroke caused by a proximal intracranial arterial occlusion in the anterior circulation, but trials that established the benefit of mechanical thrombectomy largely excluded patients with posterior circulation infarcts. However, endovascular interventions for vertebrobasilar occlusions, including mechanical thrombectomy, may be treatment options stroke centers with appropriate expertise." An UpToDate chapter on mechanical thrombectomy for acute ischemic stroke state that, "although the benefits are uncertain, mechanical thrombectomy may be a reasonable treatment option for patients with acute ischemic stroke caused by occlusion of the basilar artery, vertebral arteries, or

posterior cerebral arteries when performed at centers with appropriate expertise." The chapter noted, in particular, that there is moderate-quality evidence that mechanical thrombectomy is beneficial for patients of Chinese ancestry who can be treated within 24 hours of moderate to severe stroke (an NIHSS score ≥ 10) caused by a basilar artery occlusion if the posterior circulation ASPECTS (pc-ASPECTS) score is consistent with a limited extent of ischemia. The authors explained that the trials supporting this recommendation are not generalizable to all patients with basilar artery stroke. The authors explained that the Chinese population has higher rates of large artery intracranial atherosclerotic disease relative to other populations, and many patients in the thrombectomy groups of both trials were also treated with angioplasty and/or stenting of the basilar artery. The authors stated that larger randomized controlled trials in more diverse populations are needed to assess the efficacy of endovascular therapy for posterior circulation stroke due to large artery occlusion.

Exosome-Based Therapies

Yin et al (2024) stated that current therapeutic strategies for ischemic stroke (IS) fall short of the desired objective of neurological functional recovery; thus, there is an urgent need to develop new methods for the treatment of this condition. Exosomes are natural cell-derived vesicles that mediate signal transduction between cells under physiological and pathological conditions. They have low immunogenicity, good stability, high delivery efficiency, and the ability to cross the blood-brain barrier. These physiological properties of exosomes have the potential to lead to new break-throughs in the treatment of IS. The rapid development of nanotechnology has advanced the use of engineered exosomes, which can effectively improve targeting ability, enhance therapeutic efficacy, and minimize the dosages needed. Advances in technology have also driven clinical translational research on exosomes. These researchers described the therapeutic effects of exosomes and their positive roles in current treatment strategies for IS, including their anti-inflammation, anti-apoptosis, autophagy-regulation, angiogenesis, neurogenesis, as well as glial scar formation reduction effects. However, it is worth noting that, despite their significant therapeutic potential, there remains a dearth of standardized characterization methods and efficient isolation techniques capable of producing highly purified exosomes. These investigators stated that future optimization strategies should prioritize the examination of suitable isolation techniques and the establishment of unified work-flows to effectively harness exosomes for diagnostic or therapeutic applications in IS. The authors summarized the understanding of exosome-based treatment prospects in IS and foster innovative ideas for the development of exosome-based therapies.

Vagus Nerve Stimulation for the Improvement of Myocardial Atrophy after Ischemic Stroke

Tan et al (2024) noted that IS is one of the leading causes of chronic disability globally, and stroke-induced heart damage can result in death. According to research, patients with a variety of brain disease have good clinical results following vagus nerve stimulation (VNS). After IS, mast cells (MCs) degranulate and release a large number of mediators, which may cause systemic inflammation. Chymase secreted by MCs can increase the levels of pathological angiotensin II (Ang II), which plays an important role in the deterioration of heart disease. These researchers developed a minimally invasive, targeted, and convenient VNS approach to examine the impact of VNS and to clarify the relationship between VNS and MCs in the prognosis of patients with myocardial atrophy following AIS. They verified the role of VNS in the treatment of myocardial atrophy after stroke and its molecular mechanism using a rat model of middle cerebral artery occlusion (MCAO/r). Behavioral studies were assessed using neurobehavioral deficit scores. Enzyme-linked immunosorbent assays (ELISAs), immunofluorescence staining, Western blotting, and qRT-PCR were employed to examine the expression levels of myocardial atrophy, MC, and inflammatory markers in rat hearts. VNS improved myocardial atrophy in MCAO/r rats, inhibited MC activation, reduced the expression of chymase and Ang II, and inhibited the expression of pro-inflammatory factors. The chymase activator C48/80 reversed these effects of VNS. Chymase activation inhibited the effect of VNS on myocardial atrophy in MCAO/r rats, increased Ang II expression and aggravated inflammation and autophagy. The myocardial atrophy of MCAO/r rats was improved after chymase inhibition, and Ang II expression, inflammation and autophagy were reduced. These findings suggested that VNS may reduce the expression of chymase and Ang II by inhibiting MC activation; thus, improving myocardial atrophy and reducing inflammation and autophagy in MCAO/r rats. Inhibition of MC activation may be an effective strategy for the treatment of myocardial atrophy after stroke. The authors concluded that VNS

inhibited MC activation and reduced the expression of chymase and Ang II; thus, alleviating myocardial atrophy, inflammation and autophagy after stroke. This study demonstrated for the 1st time the regulatory effect of VNS on MC degranulation in patients with myocardial atrophy after IS, providing a new idea for the clinical treatment of myocardial atrophy after stroke.

Hypoxia-Induced Neuroprotection (Intermittent Hypoxic Exposures or Intermittent Hypoxic-Hyperoxic Exposures)

Nyamukondiwa et al (2025) noted that stroke recovery is a multi-faceted process influenced by various neuro-protective mechanisms that support long-term rehabilitation. Recent studies on hypoxia-induced neuro-protection have demonstrated promising potential in enhancing stroke recovery via adaptive cellular responses. Hypoxic conditioning in techniques of passive intermittent hypoxic exposures (IHE) or intermittent hypoxic-hyperoxic exposures (IHHE), which alternates between low and normal/high oxygen levels, is emerging as a novel complementary therapy that may improve post-stroke outcomes by promoting neuro-protection, neurogenesis, as well as vascular remodeling. These investigators examined the therapeutic implications of IHE/IHHE as a novel complementary therapy to mitigate post-stroke exacerbations and enhance recovery through various physiological and molecular mechanisms. They carried out a comprehensive literature search using public data-bases such as PubMed, Scopus, Relemed, the National Library of Medicine, and Google Scholar. The search focused on studies related to hypoxia training, neuro-protection, stroke recovery, and IHE/IHHE. The review synthesized current findings on the pathophysiological insights and therapeutic potential of intermittent hypoxic conditioning in stroke rehabilitation. These researchers highlighted several key areas where IHE/IHHE showed adaptive responses involving hypoxia-inducible factor (HIF) signaling, reactive oxygen species (ROS) regulation, and mitochondrial energetics, contributing to enhanced neuro-protection and tissue recovery. Angiogenesis and vascular remodeling: IHE/IHHE promotes angiogenesis and improves CBF, facilitating vascular remodeling and improved perfusion in damaged brain areas. Neurogenesis: IHE/IHHE enhanced neurogenesis, aiding in brain repair and functional recovery by promoting neuronal survival and regeneration. Cognitive and motor function: IHE/IHHE has been shown to improve cognitive performance and motor function in post-stroke patients, as well as in elderly individuals with mild cognitive impairment (MCI). Cardio-protection: IHE/IHHE reduced cardiovascular risk factors, such as hypertension and inflammation, and has been reported to enhance cardiac function in patients with ischemic heart disease. Integrative rehabilitation: Incorporating IHE/IHHE into post-stroke rehabilitation programs may enhance physical and cognitive outcomes, supporting a holistic approach to recovery. The authors concluded that hypoxic conditioning in modes of IHE/IHHE represents a promising complementary therapy for stroke recovery, leveraging adaptive responses to hypoxia and hyperoxia to promote neuro-protection, neurogenesis, as well as vascular remodeling. These researchers stated that further investigations with well-designed clinical trials are needed to optimize IHHE protocols, understand their long-term effects, and integrate them into clinical practice.

Nanoparticle-Based Therapies

Yang et al (2025) noted that ischemic stroke (IS) represents a significant health burden worldwide. Current therapeutic options face problems such as window narrowing and re-perfusion injury risk. Moreover, with increasing aging and risk factors, novel treatment strategies are urgently needed. NADPH oxidase (NOX)-mediated oxidative stress in microglia is an important mechanism driving neuroinflammation and cell death. Hv1, a voltage-gated proton channel highly expressed in microglia, synergizes with NOX to generate ROS, exacerbating brain damage. YHV984, a potent Hv1 inhibitor, alleviates post-IS neuroinflammation but faces clinical limitations due to potential toxic side effects and solubility issues. To improve the physico-chemical and pharmacokinetic properties of YHV984 for specific Hv1 inhibition in the brain, the multi-functional nanoparticles consisting of a T7-targeting peptide and lipid nanoparticles (LNP) were developed to deliver YHV984 (T7-LNP@YHV984). The results showed that T7-LNP@YHV984 exhibited good stability and brain targeting capability, effectively crossing the BBB and accumulating within microglia. This targeted delivery significantly suppressed Hv1 expression and activation of the NLRP3 inflammasome pathway in the damaged brain. In addition, it promoted the polarization of microglia towards the M2 phenotype, enhancing the release of anti-inflammatory factors, alleviating neuroinflammation and improved neuronal survival. Furthermore, T7-LNP@YHV984 improved survival and facilitated neurological recovery in post-IS mice. The authors concluded that T7-LNP@YHV984 multi-

functional nanoparticles with long-term stability emerged as a potent strategy to alleviate re-perfusion injury and inhibit neuroinflammation post-IS. By targeting Hv1 in microglia, the nanoparticles suppressed microglia-induced neuroinflammation, minimizing off-target effects. This innovation offers novel insights into stroke treatment and neuro-protective strategies.

Sundus et al (2025) stated that functional neurological recovery remains the primary objective when treating IS; however, current therapeutic approaches often fall short of achieving optimal outcomes. One of the most significant challenges in stroke treatment is the effective delivery of neuro-protective agents across the BBB to ischemic regions within the brain. These researchers examined the emerging role of nanoparticle-based therapies for the treatment of IS and assessed their potential to revolutionize drug delivery, enhance neuro-protection, and promote functional recovery. Recent advancements in nanotechnology have resulted in the development of engineered nanoparticles specifically designed to overcome the BBB; thereby, enabling the targeted delivery of therapeutic agents directly to the affected brain areas. Pre-clinical studies have shown the potential of nanoparticle-based therapies to activate key neuro-protective pathways, such as the phosphoinositide 3-kinase/protein kinase B/cAMP response element-binding protein signaling cascade, which is critical for neuronal survival, synaptic plasticity, as well as post-stroke recovery. By modulating these pathways, nanoparticles could mitigate neuronal damage, reduce inflammation, and promote tissue repair. In addition, nanoparticles offer a unique advantage by enabling multi-modal therapeutic strategies that simultaneously target multiple pathological mechanisms of IS, including oxidative stress, neuroinflammation, and apoptosis. This multi-faceted approach enhances the effectiveness of treatment, addressing the complex and inter-connected processes that contribute to stroke-related brain injury. Surface modifications, such as functionalization with specific ligands or targeting molecules, further improve the precision of drug delivery, enhance targeting specificity, and prolong systemic circulation; thus, optimizing therapeutic outcomes. Nanoparticle-based therapeutics represent a paradigm shift in the management of IS and provide a promising avenue for reducing post-stroke disability and improving the outcomes of long-term rehabilitation. By combining targeted drug delivery with the ability to modulate critical neuro-protective pathways, nanoparticle-based therapies hold the potential to transform the treatment landscape for IS. However, while pre-clinical data were highly encouraging, significant challenges remain in translating these advancements into clinical practice. These researchers stated that further investigations are needed to refine nanoparticle designs, optimize their safety profiles, and ensure their scalability for widespread application. The authors concluded that well-designed clinical trials are needed to validate their effectiveness, evaluate long-term biocompatibility, and address potential off-target effects. The integration of inter-disciplinary approaches, combining insights from nanotechnology, neuroscience, and pharmacology, will be crucial if researchers are to overcome these challenges. Ultimately, nanoparticle-based therapies offer a foundation for innovative, precision-based treatments that could improve outcomes for IS patients; therefore, paving the way for a new era in stroke care and neurological rehabilitation.

Li et al (2025) noted that IS poses a significant threat to public health, and current therapeutic effectiveness is limited by BBB permeability and multi-factorial pathophysiology (such as oxidative stress and inflammation). In this study, a self-assembled metal-organic framework (Zn-DB) was decorated with CeO₂ nanoparticles to construct the CZB nano-composite with anti-oxidative and anti-apoptotic properties. The CZB-D-A multi-functional nano-composite was developed by co-loading deferoxamine and an anti-C5a aptamer, and CZB-D-A was camouflaged with neutrophil-like membranes modified with RVG29 to synthesize a biomimetic nano-platform (RH@CZB-D-A) for precise targeting and penetration of the BBB as well as immune evasion. Subsequently, this nano-platform precisely delivered therapeutic agents and released them better in the acidic lesion owing to its pH-acid response. It was designed to depolarize microglia from the destructive M1 phenotype to the protective M2 phenotype and alleviate ischemia-reperfusion injury by suppressing neuroinflammation, scavenging ROS, and mitigating oxidative stress. Meanwhile, it synergistically promoted neuro-protection by multiple mechanisms of anti-oxidation, anti-apoptosis, anti-ferroptosis, and anti-inflammation, both in-vitro and in-vivo. Furthermore, administration of RH@CZB-D-A to MCAO mice not only enhanced neuronal regeneration but also improved neurological deficits and promoted spatial memory and learning ability. These researchers stated that the findings of this study could bring an alternative approach for developing promising therapeutic strategies against IS for clinical applications.

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The above policy is based on the following references:

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